

MODERN INSTITUTE OF TECHNOLOGY AND RESEARCH CENTRE, ALWAR

Name of faculty: Sunil Kumar Sharma

Subject & Subject Code: AEM, 3CS2-01, 3AD1-01

Semester: III

Session: 2022-23(Odd Sem.)

Branch: CSE, AIDS

Batch: A, B & C

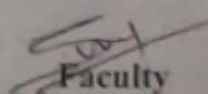
Unit No.2

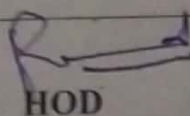
Date of Submission: 18-10-2022

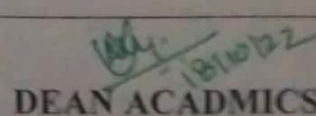
Lecture No.	Topics Covered	Page No.	Total Page
7	Binomial Distribution	113-116	4
8	Poisson Distribution	117-119	3
9	Normal Distribution	120-125	6
10	Uniform Distribution	126-128	3
11	Exponential Distribution	129-139	3
12	Curve Fitting	140-142	11
13	Correlation: Correlation Coefficient	143-146	3
14	Karl Pearson Pearson's Correlation Coefficient	147-155	4
15	Spearsman's Rank Correlation Coefficient	156-161	9
16	Line of Regression and regression coefficients	165-167	6
17	Angle between lines of regression	168-170	3

References:

Chandrika Prasad, Advance Engineering Mathematics, Stanford Publishers and Distributors
Engineering Mathematics-T. Veerarajan, TMH


Faculty


HOD


DEAN ACADMICS

UNIT-2 (ADVANCE ENGINEERING MATHEMATICS)

BINOMIAL DISTRIBUTION

LECTURE NO: -7

Binomial Distribution:

A **binomial experiment** is a statistical experiment that has the following properties:

- The experiment consists of n repeated trials.
- Each trial can result in just two possible outcomes. We call one of these outcomes a success and the other, a failure.
- The probability of success, denoted by P , is the same on every trial.
- The trials are independent; that is, the outcome on one trial does not affect the outcome on other trials.

Consider the following statistical experiment. You flip a coin 2 times and count the number of times the coin lands on heads. This is a binomial experiment because:

- The experiment consists of repeated trials. We flip a coin 2 times.
- Each trial can result in just two possible outcomes - heads or tails.
- The probability of success is constant - 0.5 on every trial.
- The trials are independent; that is, getting heads on one trial does not affect whether we get heads on other trials.

Notation

The following notation is helpful, when we talk about binomial probability.

- x : The number of successes that result from the binomial experiment.
- n : The number of trials in the binomial experiment.
- P : The probability of success on an individual trial.
- Q : The probability of failure on an individual trial. (This is equal to $1 - P$.)
- $n!$: The [factorial](#) of n (also known as n factorial).
- $b(x; n, P)$: Binomial probability - the probability that an n -trial binomial experiment results in exactly x successes, when the probability of success on an individual trial is P .
- ${}_nC_r$: The number of [combinations](#) of n things, taken r at a time.

Binomial Distribution

A **binomial random variable** is the number of successes x in n repeated trials of a binomial experiment. The [probability distribution](#) of a binomial random variable is called a **binomial distribution**.

Suppose we flip a coin two times and count the number of heads (successes). The binomial random variable is the number of heads, which can take on values of 0, 1, or 2. The binomial distribution is presented below.

Number of heads	Probability
0	0.25
1	0.50
2	0.25

The binomial distribution has the following properties:

- The mean of the distribution (μ_x) is equal to $n * P$.
- The [variance](#) (σ^2_x) is $n * P * (1 - P)$.
- The [standard deviation](#) (σ_x) is $\sqrt{n * P * (1 - P)}$.
- **Binomial Formula and Binomial Probability**
- The **binomial probability** refers to the probability that a binomial experiment results in exactly x successes. For example, in the above table, we see that the binomial probability of getting exactly one head in two coin flips is 0.50.
- Given x , n , and P , we can compute the binomial probability based on the binomial formula:

- **Binomial Formula.** Suppose a binomial experiment consists of n trials and results in x successes. If the probability of success on an individual trial is P , then the binomial probability is:

$$b(x; n, P) = {}_n C_x * P^x * (1 - P)^{n-x}$$

or

$$b(x; n, P) = \{ n! / [x! (n - x)!] \} * P^x * (1 - P)^{n-x}$$

▪

▪ Example 1

Suppose a die is tossed 5 times. What is the probability of getting exactly 2 fours?

- **Solution:** This is a binomial experiment in which the number of trials is equal to 5, the number of successes is equal to 2, and the probability of success on a single trial is $1/6$ or about 0.167. Therefore, the binomial probability is:

$$b(2; 5, 0.167) = {}_5C_2 * (0.167)^2 * (0.833)^3$$

$$b(2; 5, 0.167) = 0.161$$

▪ Cumulative Binomial Probability

- A **cumulative binomial probability** refers to the probability that the binomial random variable falls within a specified range (e.g., is greater than or equal to a stated lower limit and less than or equal to a stated upper limit).
- For example, we might be interested in the cumulative binomial probability of obtaining 45 or fewer heads in 100 tosses of a coin (see Example 1 below). This would be the sum of all these individual binomial probabilities.

$$b(x \leq 45; 100, 0.5) =$$

$$b(x = 0; 100, 0.5) + b(x = 1; 100, 0.5) + \dots + b(x = 44; 100, 0.5) + b(x = 45; 100, 0.5)$$

▪ Binomial Calculator

- As you may have noticed, the binomial formula requires many time-consuming computations. The Binomial Calculator can do this work for you - quickly, easily, and error-free. Use the Binomial Calculator to compute binomial probabilities and cumulative binomial probabilities. The calculator is free. It can found in the Stat Trek main menu under the Stat Tools tab. Or you can tap the button below.
- [Binomial Calculator](#)

▪ Example 2

What is the probability of obtaining 45 or fewer heads in 100 tosses of a coin?

- *Solution:* To solve this problem, we compute 46 individual probabilities, using the binomial formula. The sum of all these probabilities is the answer we seek. Thus,
- $b(x \leq 45; 100, 0.5) = b(x = 0; 100, 0.5) + b(x = 1; 100, 0.5) + \dots + b(x = 45; 100, 0.5)$

$$b(x \leq 45; 100, 0.5) = 0.184$$

▪ Example 3

The probability that a student is accepted to a prestigious college is 0.3. If 5 students from the same school apply, what is the probability that at most 2 are accepted?

- *Solution:* To solve this problem, we compute 3 individual probabilities, using the binomial formula. The sum of all these probabilities is the answer we seek. Thus,

$$b(x \leq 2; 5, 0.3) = b(x = 0; 5, 0.3) + b(x = 1; 5, 0.3) + b(x = 2; 5, 0.3)$$

$$b(x \leq 2; 5, 0.3) = 0.1681 + 0.3601 + 0.3087$$

$$b(x \leq 2; 5, 0.3) = 0.8369$$

▪ Example 4

What is the probability that the world series will last 4 games? 5 games? 6 games? 7 games? Assume that the teams are evenly matched.

- *Solution:* This is a very tricky application of the binomial distribution. If you can follow the logic of this solution, you have a good understanding of the material covered in the tutorial, to this point.
- In the world series, there are two baseball teams. The series ends when the winning team wins 4 games. Therefore, we define a success as a win by the team that ultimately becomes the world series champion.
- For the purpose of this analysis, we assume that the teams are evenly matched. Therefore, the probability that a particular team wins a particular game is 0.5.
- Let's look first at the simplest case. What is the probability that the series lasts only 4 games. This can occur if one team wins the first 4 games. The probability of the National League team winning 4 games in a row is:
 - $b(4; 4, 0.5) = {}_4C_4 * (0.5)^4 * (0.5)^0 = 0.0625$
- Similarly, when we compute the probability of the American League team winning 4 games in a row, we find that it is also 0.0625. Therefore, probability that the series ends in four games would be $0.0625 + 0.0625 = 0.125$; since the series would end if either the American or National League team won 4 games in a row.
- Now let's tackle the question of finding probability that the world series ends in 5 games. The trick in finding this solution is to recognize that the series can only end in 5 games, if one team has won 3 out of the first 4 games. So let's first find the probability that the American League team wins exactly 3 of the first 4 games.
 - $b(3; 4, 0.5) = {}_4C_3 * (0.5)^3 * (0.5)^1 = 0.25$
- Okay, here comes some more tricky stuff, so listen up. Given that the American League team has won 3 of the first 4 games, the American League team has a 50/50 chance of winning the fifth game to end the series. Therefore, the probability of the American League team winning the series in 5 games is $0.25 * 0.50 = 0.125$. Since the National League team could also win the series in 5 games, the probability that the series ends in 5 games would be $0.125 + 0.125 = 0.25$.
- The rest of the problem would be solved in the same way. You should find that the probability of the series ending in 6 games is 0.3125; and the probability of the series ending in 7 games is also 0.3125.

References:

Advance Engineering Mathematics, Chandrika Prasad, Standard Publishers and Distributors

UNIT-2 (ADVANCE ENGINEERING MATHEMATICS)

POISSON DISTRIBUTION

LECTURE NO: -8

Poisson Distribution:

A Poisson distribution is the probability distribution that results from a Poisson experiment.

Attributes of a Poisson Experiment

A **Poisson experiment** is a statistical experiment that has the following properties:

- The experiment results in outcomes that can be classified as successes or failures.
- The average number of successes (μ) that occurs in a specified region is known.
- The probability that a success will occur is proportional to the size of the region.
- The probability that a success will occur in an extremely small region is virtually zero.

Note that the specified region could take many forms. For instance, it could be a length, an area, a volume, a period of time, etc.

Notation

The following notation is helpful, when we talk about the Poisson distribution.

- e : A constant equal to approximately 2.71828. (Actually, e is the base of the natural logarithm system.)
- μ : The mean number of successes that occur in a specified region.
- x : The actual number of successes that occur in a specified region.
- $P(x; \mu)$: The **Poisson probability** that exactly x successes occur in a Poisson experiment, when the mean number of successes is μ .

Poisson Distribution

A **Poisson random variable** is the number of successes that result from a Poisson experiment. The probability distribution of a Poisson random variable is called a **Poisson distribution**.

Given the mean number of successes (μ) that occur in a specified region, we can compute the Poisson probability based on the following formula:

Poisson Formula. Suppose we conduct a Poisson experiment, in which the average number of successes within a given region is μ . Then, the Poisson probability is:

$$P(x; \mu) = (e^{-\mu}) (\mu^x) / x!$$

where x is the actual number of successes that result from the experiment, and e is approximately equal to 2.71828.

The Poisson distribution has the following properties:

- The mean of the distribution is equal to μ .
- The variance is also equal to μ .

Poisson Distribution Example

The average number of homes sold by the Acme Realty company is 2 homes per day. What is the probability that exactly 3 homes will be sold tomorrow?

Solution: This is a Poisson experiment in which we know the following:

- $\mu = 2$; since 2 homes are sold per day, on average.
- $x = 3$; since we want to find the likelihood that 3 homes will be sold tomorrow.
- $e = 2.71828$; since e is a constant equal to approximately 2.71828.

We plug these values into the Poisson formula as follows:

$$P(x; \mu) = (e^{-\mu}) (\mu^x) / x!$$

$$P(3; 2) = (2.71828^{-2}) (2^3) / 3!$$

$$P(3; 2) = (0.13534) (8) / 6$$

$$P(3; 2) = 0.180$$

Thus, the probability of selling 3 homes tomorrow is 0.180.

Poisson Calculator

Clearly, the Poisson formula requires many time-consuming computations. The Stat Trek Poisson Calculator can do this work for you - quickly, easily, and error-free. Use the

Poisson Calculator to compute Poisson probabilities and cumulative Poisson probabilities. It can found in the Stat Trek main menu under the Stat Tools tab. Or you can tap the button below.

Poisson Calculator

Cumulative Poisson Probability

A **cumulative Poisson probability** refers to the probability that the Poisson random variable is greater than some specified lower limit and less than some specified upper limit.

Cumulative Poisson Example

Suppose the average number of lions seen on a 1-day safari is 5. What is the probability that tourists will see fewer than four lions on the next 1-day safari?

Solution: This is a Poisson experiment in which we know the following:

- $\mu = 5$; since 5 lions are seen per safari, on average.
- $x = 0, 1, 2, \text{ or } 3$; since we want to find the likelihood that tourists will see fewer than 4 lions; that is, we want the probability that they will see 0, 1, 2, or 3 lions.
- $e = 2.71828$; since e is a constant equal to approximately 2.71828.

To solve this problem, we need to find the probability that tourists will see 0, 1, 2, or 3 lions. Thus, we need to calculate the sum of four probabilities: $P(0; 5) + P(1; 5) + P(2; 5) + P(3; 5)$. To compute this sum, we use the Poisson formula:

$$P(x \leq 3, 5) = P(0; 5) + P(1; 5) + P(2; 5) + P(3; 5)$$

$$P(x \leq 3, 5) = [(e^{-5})(5^0) / 0!] + [(e^{-5})(5^1) / 1!] + [(e^{-5})(5^2) / 2!] + [(e^{-5})(5^3) / 3!]$$

$$P(x \leq 3, 5) = [(0.006738)(1) / 1] + [(0.006738)(5) / 1] + [(0.006738)(25) / 2] + [(0.006738)(125) / 6]$$

$$P(x \leq 3, 5) = [0.0067] + [0.03369] + [0.084224] + [0.140375]$$

$$P(x \leq 3, 5) = 0.2650$$

Thus, the probability of seeing at no more than 3 lions is 0.2650.

References:

Advance Engineering Mathematics, Chandrika Prasad, Standard Publishers and Distributors

Notes By: -Sunil kumar Sharma (Assistant Professor-Maths Department)

UNIT-2 (ADVANCE ENGINEERING MATHEMATICS)

NORMAL DISTRIBUTION

LECTURE NO: -9

Normal Distribution:

The **normal distribution** refers to a family of **continuous probability distributions** described by the normal equation.

The Normal Equation

The normal distribution is defined by the following equation:

The Normal Equation. The value of the random variable Y is:

$$Y = \left\{ \frac{1}{\sigma \sqrt{2\pi}} \right\} * e^{-(x - \mu)^2 / 2\sigma^2}$$

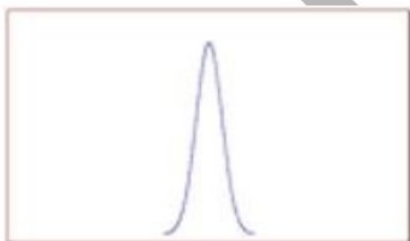
where X is a normal random variable, μ is the mean, σ is the standard deviation, π is approximately 3.14159, and e is approximately 2.71828.

The random variable X in the normal equation is called the **normal random variable**. The normal equation is the **probability density function** for the normal distribution.

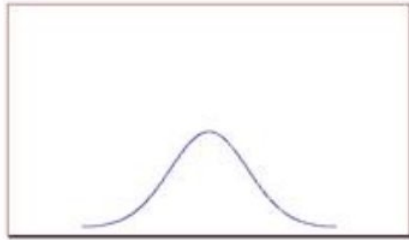
The Normal Curve

The graph of the normal distribution depends on two factors - the mean and the standard deviation. The mean of the distribution determines the location of the center of the graph, and the standard deviation determines the height and width of the graph. All normal distributions look like a symmetric, bell-shaped curve, as shown below.

Smaller standard deviation



Bigger standard deviation

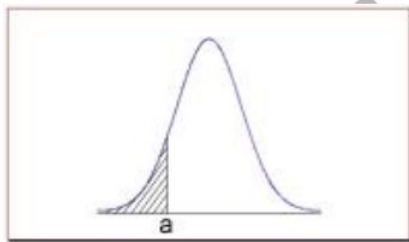


When the standard deviation is small, the curve is tall and narrow; and when the standard deviation is big, the curve is short and wide (see above)

Probability and the Normal Curve

The normal distribution is a continuous probability distribution. This has several implications for probability.

- The total area under the normal curve is equal to 1.
- The probability that a normal random variable X equals any particular value is 0.
- The probability that X is greater than a equals the area under the normal curve bounded by a and plus infinity (as indicated by the *non-shaded* area in the figure below).
- The probability that X is less than a equals the area under the normal curve bounded by a and minus infinity (as indicated by the *shaded* area in the figure below).

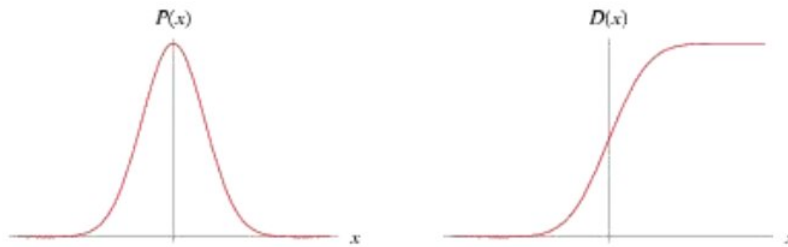


Additionally, every normal curve (regardless of its mean or standard deviation) conforms to the following "rule".

- About 68% of the area under the curve falls within 1 standard deviation of the mean.
- About 95% of the area under the curve falls within 2 standard deviations of the mean.
- About 99.7% of the area under the curve falls within 3 standard deviations of the mean.

Collectively, these points are known as the **empirical rule** or the **68-95-99.7 rule**. Clearly, given a normal distribution, most outcomes will be within 3 standard deviations of the mean.

To find the probability associated with a normal random variable, use a graphing calculator, an online normal distribution calculator, or a normal distribution table. In the examples below, we illustrate the use of Stat Trek's [Normal Distribution Calculator](#), a free tool available on this site. In the next lesson, we demonstrate the use of normal distribution tables.



A normal distribution in a variate X with mean μ and variance σ^2 is a statistic distribution with probability density function

$$P(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)} \quad (1)$$

on the domain $x \in (-\infty, \infty)$. While statisticians and mathematicians uniformly use the term "normal distribution" for this distribution, physicists sometimes call it a Gaussian distribution and, because of its curved flaring shape, social scientists refer to it as the "bell curve." Feller (1968) uses the symbol $\varphi(x)$ for $P(x)$ in the above equation, but then switches to $n(x)$ in Feller (1971).

de Moivre developed the normal distribution as an approximation to the binomial distribution, and it was subsequently used by Laplace in 1783 to study measurement errors and by Gauss in 1809 in the analysis of astronomical data (Havil 2003, p. 157).

The normal distribution is implemented in the Wolfram Language as `NormalDistribution[mu, sigma]`.

The so-called "standard normal distribution" is given by taking $\mu = 0$ and $\sigma^2 = 1$ in a general normal distribution. An arbitrary normal distribution can be converted to a standard normal distribution by changing variables to $Z = (X - \mu)/\sigma$, so $dz = dx/\sigma$, yielding

$$P(x) dx = \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz. \quad (2)$$

The Fisher-Behrens problem is the determination of a test for the equality of means for two normal distributions with different variances.

The normal distribution function $\Phi(z)$ gives the probability that a standard normal variate assumes a value in the interval $[0, z]$,

$$\Phi(z) \equiv \frac{1}{\sqrt{2\pi}} \int_0^z e^{-x^2/2} dx \quad (3)$$

$$= \frac{1}{2} \operatorname{erf}\left(\frac{z}{\sqrt{2}}\right), \quad (4)$$

where `erf` is a function sometimes called the error function. Neither $\Phi(z)$ nor `erf` can be expressed in terms of finite additions, subtractions, multiplications, and root extractions, and so both must be either computed numerically or otherwise approximated.

Problems on Normal Distribution:

There are many problems that are solved by Normal Distribution. The problems of Normal distribution are solved by using Normal Table.

1. X is a normally normally distributed variable with mean $\mu = 30$ and standard deviation $\sigma = 4$. Find
 - a) $P(x < 40)$
 - b) $P(x > 21)$
 - c) $P(30 < x < 35)$

Sol: a) For $x = 40$, the z-value $z = (40 - 30) / 4 = 2.5$

Hence $P(x < 40) = P(z < 2.5) = [\text{area to the left of } 2.5] = 0.9938$

b) For $x = 21$, $z = (21 - 30) / 4 = -2.25$

Hence $P(x > 21) = P(z > -2.25) = [\text{total area}] - [\text{area to the left of } -2.25]$

$$= 1 - 0.0122 = 0.9878$$

c) For $x = 30$, $z = (30 - 30) / 4 = 0$ and for $x = 35$, $z = (35 - 30) / 4 = 1.25$

Hence $P(30 < x < 35) = P(0 < z < 1.25) = [\text{area to the left of } z = 1.25] - [\text{area to the left of } 0]$

$$= 0.8944 - 0.5 = 0.3944$$

2. A radar unit is used to measure speeds of cars on a motorway. The speeds are normally distributed with a mean of 90 km/hr and a standard deviation of 10 km/hr. What is the probability that a car picked at random is travelling at more than 100 km/hr?

Sol: Let x be the random variable that represents the speed of cars. x has $\mu = 90$ and $\sigma = 10$.

We have to find the probability that x is higher than 100 or $P(x > 100)$

For $x = 100$, $z = (100 - 90) / 10 = 1$

$P(x > 90) = P(z > 1) = [\text{total area}] - [\text{area to the left of } z = 1]$

$$= 1 - 0.8413 = 0.1587$$

The probability that a car selected at a random has a speed greater than 100 km/hr is equal to 0.1587

3. For a certain type of computers, the length of time between charges of the battery is normally distributed with a mean of 50 hours and a standard deviation of 15 hours. John owns one of these computers and wants to know the probability that the length of time will be between 50 and 70 hours.

Sol: Let x be the random variable that represents the length of time. It has a mean of 50 and a standard deviation of 15. We have to find the probability that x is between 50 and 70 or $P(50 < x < 70)$

$$\text{For } x = 50, z = (50 - 50) / 15 = 0$$

$$\text{For } x = 70, z = (70 - 50) / 15 = 1.33 \text{ (rounded to 2 decimal places)}$$

$$\begin{aligned} P(50 < x < 70) &= P(0 < z < 1.33) = [\text{area to the left of } z = 1.33] - [\text{area to the left of } z = 0] \\ &= 0.9082 - 0.5 = 0.4082 \end{aligned}$$

The probability that John's computer has a length of time between 50 and 70 hours is equal to 0.4082.

4. Entry to a certain University is determined by a national test. The scores on this test are normally distributed with a mean of 500 and a standard deviation of 100. Tom wants to be admitted to this university and he knows that he must score better than at least 70% of the students who took the test. Tom takes the test and scores 585. Will he be admitted to this university?

Sol: Let x be the random variable that represents the scores. x is normally distributed with a mean of 500 and a standard deviation of 100. The total area under the normal curve represents the total number of students who took the test. If we multiply the values of the areas under the curve by 100, we obtain percentages.

$$\text{For } x = 585, z = (585 - 500) / 100 = 0.85$$

The proportion P of students who scored below 585 is given by

$$P = [\text{area to the left of } z = 0.85] = 0.8023 = 80.23\%$$

Tom scored better than 80.23% of the students who took the test and he will be admitted to this University.

5. The length of similar components produced by a company are approximated by a normal distribution model with a mean of 5 cm and a standard deviation of 0.02 cm. If a component is chosen at random

a) what is the probability that the length of this component is between 4.98 and 5.02 cm?

b) what is the probability that the length of this component is between 4.96 and 5.04 cm

Sol: a) $P(4.98 < x < 5.02) = P(-1 < z < 1)$

$$= 0.6826$$

b) $P(4.96 < x < 5.04) = P(-2 < z < 2)$

$$= 0.9544$$

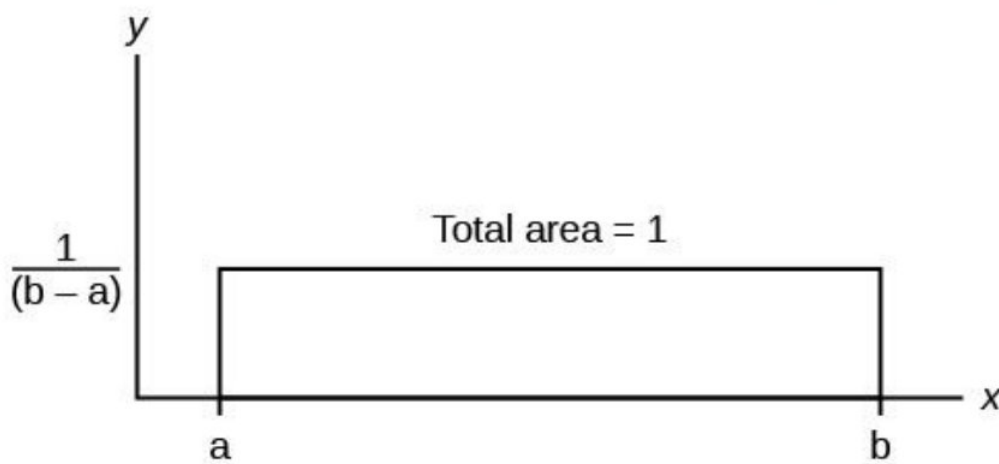
UNIT-2 (ADVANCE ENGINEERING MATHEMATICS)

THE UNIFORM DISTRIBUTION

LECTURE NO: -10

The Uniform Distribution:

If X has a uniform distribution where $a < x < b$ or $a \leq x \leq b$, then X takes on values between a and b (may include a and b). All values x are equally likely. We write $X \sim U(a, b)$. The mean of X is $\mu = a + b/2$. X is continuous.



The probability $P(c < X < d)$ may be found by computing the area under $f(x)$, between c and d . Since the corresponding area is a rectangle, the area may be found simply by multiplying the width and the height.

Formula Review

X = a real number between a and b (in some instances, X can take on the values a and b). a = smallest X ; b = largest X

$$X \sim U(a, b)$$

The mean is $\mu = a + b/2$ The standard deviation is $\sigma = \sqrt{(b-a)/12}$

Probability density function: $f(x) = \frac{1}{b-a}$ for $a \leq x \leq b$

Area to the Left of x : $P(X < x) = (x-a)(\frac{1}{b-a})$

Area to the Right of x : $P(X > x) = (b-x)(\frac{1}{b-a})$

Area Between c and d : $P(c < x < d) = (\text{base})(\text{height}) = (d-c)(\frac{1}{b-a})$

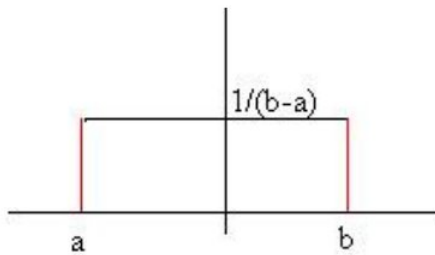
Uniform: $X \sim U(a, b)$ where $a < x < b$

- pdf: $f(x) = \frac{1}{b-a}$ for $a \leq x \leq b$
- cdf: $P(X \leq x) = \frac{x-a}{b-a}$
- mean: $\mu = a + \frac{b-a}{2}$
- standard deviation: $\sigma = \sqrt{\frac{(b-a)^2}{12}}$
- $P(c < X < d) = (d - c)$

A continuous random variable X which has probability density function given by:

$$f(x) = \frac{1}{b-a} \text{ for } a \leq x \leq b$$

(and $f(x) = 0$ if x is not between a and b) follows a **uniform distribution** with parameters a and b . We write $X \sim U(a, b)$



Remember that the area under the graph of the random variable must be equal to 1 (see continuous random variables).

Expectation and Variance

If $X \sim U(a, b)$, then:

- $E(X) = \frac{1}{2}(a + b)$
- $\text{Var}(X) = \frac{1}{12}(b - a)^2$

Proof of Expectation

$$\int_a^b \frac{x}{b-a} dx = \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b = \frac{b^2 - a^2}{2(b-a)} = \frac{b+a}{2}$$

Cumulative Distribution Function

The cumulative distribution function can be found by integrating the p.d.f between 0 and t :

$$F(t) = \int_a^t \frac{1}{b-a} dx = \frac{t-a}{b-a}$$

The uniform distribution is a continuous probability distribution and is concerned with events that are equally likely to occur. When working out problems that have a uniform distribution, be careful to note if the data is inclusive or exclusive.

Example 1

The data in the table below are 55 smiling times, in seconds, of an eight-week-old baby.

10.4	19.6	18.8	13.9	17.8	16.8	21.6	17.9
12.8	14.8	22.8	20.0	15.9	16.3	13.4	17.1
1.3	0.7	8.9	11.9	10.9	7.3	5.9	3.7
5.8	6.9	2.6	5.8	21.7	11.8	3.4	2.1
8.9	9.4	9.4	7.6	10.0	3.3	6.7	7.8

The sample mean = 11.49 and the sample standard deviation = 6.23.

We will assume that the smiling times, in seconds, follow a uniform distribution between zero and 23 seconds, inclusive. This means that any smiling time from zero to and including 23 seconds is equally likely. The histogram that could be constructed from the sample is an empirical distribution that closely matches the theoretical uniform distribution.

Let X = length, in seconds, of an eight-week-old baby's smile.

The notation for the uniform distribution is $X \sim U(a, b)$ where a = the lowest value of x and b = the highest value of x .

The probability density function is $f(x) = 1/(b-a)$ for $a \leq x \leq b$.

For this example, $X \sim U(0, 23)$ and $f(x) = 1/23 - 0$ for $0 \leq X \leq 23$.

Formulas for the theoretical mean and standard deviation are $\mu = a + b/2$ and $\sigma = \sqrt{(b-a)/12}$

For this problem, the theoretical mean and standard deviation are $\mu = 0 + 23/2 = 11.50$ seconds and $\sigma = \sqrt{(23-0)/12} = 6.64$ seconds

Notice that the theoretical mean and standard deviation are close to the sample mean and standard deviation in this example.

References:

Advance Engineering Mathematics, Chandrika Prasad, Standard Publishers and Distributors

UNIT-2 (ADVANCE ENGINEERING MATHEMATICS)

EXPONENTIAL DISTRIBUTION

LECTURE NO: -11

Exponential Distribution:

The **exponential distribution** is often concerned with the amount of time until some specific event occurs. For example, the amount of time (beginning now) until an earthquake occurs has an exponential distribution. Other examples include the length, in minutes, of long distance business telephone calls, and the amount of time, in months, a car battery lasts. It can be shown, too, that the value of the change that you have in your pocket or purse approximately follows an exponential distribution.

Values for an exponential random variable occur in the following way. There are fewer large values and more small values. For example, the amount of money customers spend in one trip to the supermarket follows an exponential distribution. There are more people who spend small amounts of money and fewer people who spend large amounts of money.

The exponential distribution is widely used in the field of reliability. Reliability deals with the amount of time a product lasts.

EXAMPLE

Let X = amount of time (in minutes) a postal clerk spends with his or her customer. The time is known to have an exponential distribution with the average amount of time equal to four minutes.

X is a **continuous random variable** since time is measured. It is given that $\mu = 4$ minutes. To do any calculations, you must know m , the decay parameter.

$m = 1/\mu = 1/4$. Therefore, $m = 1/4 = 0.25$.

The standard deviation, σ , is the same as the mean. $\mu = \sigma$

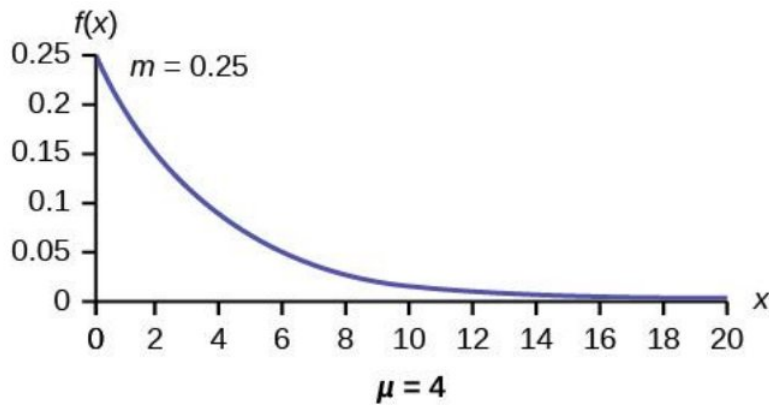
The distribution notation is $X \sim \text{Exp}(m)$. Therefore, $X \sim \text{Exp}(0.25)$.

The probability density function is $f(x) = me^{-mx}$. The number $e = 2.71828182846...$ It is a number that is used often in mathematics. Scientific calculators have the key " e^x ." If you enter one for x , the calculator will display the value e .

The curve is:

$f(x) = 0.25e^{-0.25x}$ where x is at least zero and $m = 0.25$.

For example, $f(5) = 0.25e^{-(0.25)(5)} = 0.072$. The postal clerk spends five minutes with the customers. The graph is as follows:



Notice the graph is a declining curve. When $x = 0$, $f(x) = 0.25e^{(-0.25)(0)} = (0.25)(1) = 0.25 = m$. The maximum value on the y -axis is m .

EXAMPLE

Using the information in example 1, find the probability that a clerk spends four to five minutes with a randomly selected customer.

The curve is:

$$X \sim \text{Exp}(0.125); f(x) = 0.125e^{-0.125x}$$

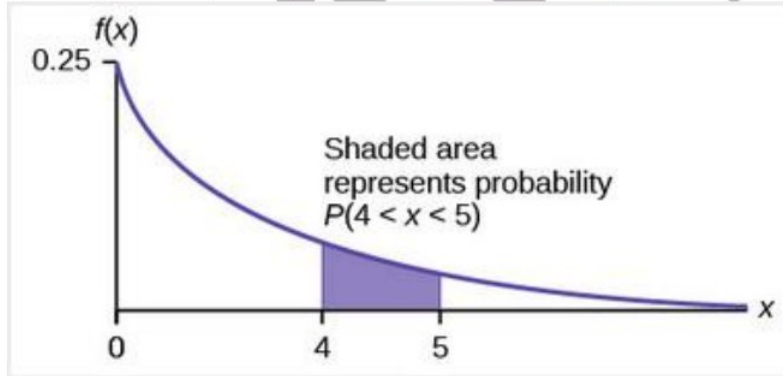
a) Find $P(4 < x < 5)$.

Solution:

The cumulative distribution function (CDF) gives the area to the left.

$$P(x < x) = 1 - e^{-mx}$$

$$P(x < 5) = 1 - e^{(-0.25)(5)} = 0.7135 \text{ and } P(x < 4) = 1 - e^{(-0.25)(4)} = 0.6321$$



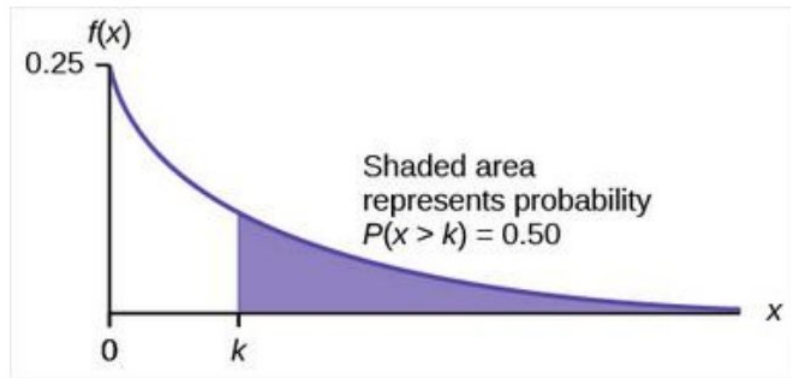
You can do these calculations easily on a calculator.

The probability that a postal clerk spends four to five minutes with a randomly selected customer is $P(4 < x < 5) = P(x < 5) - P(x < 4) = 0.7135 - 0.6321 = 0.0814$.

On the home screen, enter $(1 - e^{(-0.25*5)}) - (1 - e^{(-0.25*4)})$ or enter $e^{(-0.25*4)} - e^{(-0.25*5)}$. b) Half of all customers are finished within how long? (Find the 50th percentile)

Solution:

Find the 50th percentile.



$P(x < k) = 0.50$, $k = 2.8$ minutes (calculator or computer)
Half of all customers are finished within 2.8 minutes.

You can also do the calculation as follows:

$$P(x < k) = 0.50 \text{ and } P(x < k) = 1 - e^{-0.25k}$$

$$\text{Therefore, } 0.50 = 1 - e^{-0.25k} \text{ and } e^{-0.25k} = 1 - 0.50 = 0.5$$

$$\text{Take natural logs: } \ln(e^{-0.25k}) = \ln(0.50). \text{ So, } -0.25k = \ln(0.50)$$

$$\text{Solve for } k: k = \ln(0.50) / -0.25 = 2.8 \text{ minutes}$$

c) Which is larger, the mean or the median?

Solution:

From part b, the median or 50th percentile is 2.8 minutes. The theoretical mean is four minutes. The mean is larger.

EXAMPLE

The number of days ahead travelers purchase their airline tickets can be modeled by an exponential distribution with the average amount of time equal to 15 days. Find the probability that a traveler will purchase a ticket fewer than ten days in advance. How many days do half of all travelers wait?

Solution:

$$P(x < 10) = 0.4866$$

$$50^{\text{th}} \text{ percentile} = 10.40$$

References:

Advance Engineering Mathematics, Chandrika Prasad, Standard Publishers and Distributors

UNIT-2 (ADVANCE ENGINEERING MATHEMATICS)

LEAST SQUARE METHOD-CURVE FITTING

LECTURE NO: -12

Least Square Method- Curve Fitting:

The least-squares method is a crucial statistical method that is practised to find a regression line or a best-fit line for the given pattern. This method is described by an equation with specific parameters. The method of least squares is generously used in evaluation and regression. In regression analysis, this method is said to be a standard approach for the approximation of sets of equations having more equations than the number of unknowns.

The method of least squares actually defines the solution for the minimization of the sum of squares of deviations or the errors in the result of each equation. Find the formula for sum of squares of errors, which help to find the variation in observed data.

The least-squares method is often applied in data fitting. The best fit result is assumed to reduce the sum of squared errors or residuals which are stated to be the differences between the observed or experimental value and corresponding fitted value given in the model.

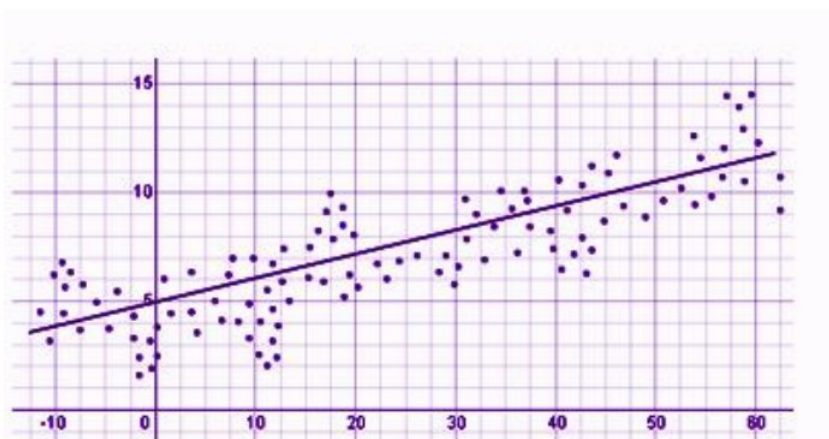
There are two basic categories of least-squares problems:

- Ordinary or linear least squares
- Nonlinear least squares

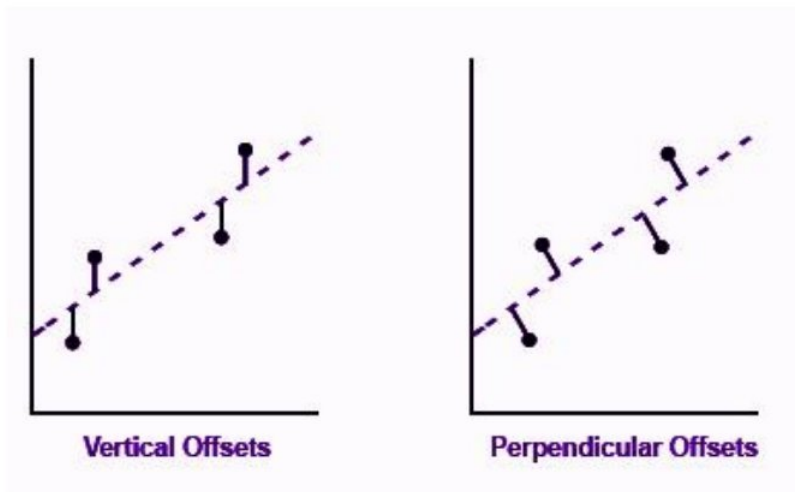
These depend upon linearity or nonlinearity of the residuals. The linear problems are often seen in regression analysis in statistics. On the other hand, the non-linear problems generally used in the iterative method of refinement in which the model is approximated to the linear one with each iteration.

Least Square Method Graph

In linear regression, the line of best fit is a straight line as shown in the following diagram:



The given data points are to be minimized by the method of reducing residuals or offsets of each point from the line. The vertical offsets are generally used in surface, polynomial and hyperplane problems, while perpendicular offsets are utilized in common practice.



Least Square Method Formula

The least-square method states that the curve that best fits a given set of observations, is said to be a curve having a minimum sum of the squared residuals (or deviations or errors) from the given data points. Let us assume that the given points of data are (x_1, y_1) , (x_2, y_2) , (x_3, y_3) , ..., (x_n, y_n) in which all x 's are independent variables, while all y 's are dependent ones. Also, suppose that $f(x)$ be the fitting curve and d represents error or deviation from each given point.

Now, we can write:

$$d_1 = y_1 - f(x_1)$$

$$d_2 = y_2 - f(x_2)$$

$$d_3 = y_3 - f(x_3)$$

.....

$$d_n = y_n - f(x_n)$$

The least-squares explain that the curve that best fits is represented by the property that the sum of squares of all the deviations from given values must be minimum. I.e:

$$S = \sum_{i=1}^n d_i^2$$

$$S = \sum_{i=1}^n [y_i - f_{x_i}]^2$$

$$S = d_1^2 + d_2^2 + d_3^2 + \dots + d_n^2$$

Sum = Minimum Quantity

Limitations for Least-Square Method

The least-squares method is a very beneficial method of curve fitting. Despite many benefits, it has a few shortcomings too. One of the main limitations is discussed here.

In the process of regression analysis, which utilizes the least-square method for curve fitting, it is inevitably assumed that the errors in the independent variable are negligible or zero. In such

cases, when independent variable errors are non-negligible, the models are subjected to measurement errors. Therefore, here, the least square method may even lead to hypothesis testing, where parameter estimates and confidence intervals are taken into consideration due to the presence of errors occurring in the independent variables.

There are a total of N observations of a y -value for the input x . If we assume the data may be represented mathematically by an equation for a straight line, we write

$$y = a + bx$$

Equation 1

By carefully selecting the two parameters a and b , we may find an equation which closely imitates the relationship between y and x portrayed in Figure 1.

Table 1. A small linear dataset used to construct the previous figure.

i	x	y
1	10	0.51
2	20	0.69
3	30	0.92
4	40	1.21
5	50	1.33
6	60	1.61
7	70	1.76
8	80	2
:	:	:
N	x_N	y_N

Unfortunately, for any given input x_i , the observation y_i is not necessarily exactly the value predicted by Equation 1. The error, ε_i , is therefore given as

$$\varepsilon_i = y_i - (a + bx_i)$$

Equation 2

In an effort to find the **best** values of a, b which minimize the errors, we might take a derivative of the total error (summation of all the ε_i) with respect to a and b , set them equal to zero, and solve for the roots of these simultaneous equations. Unfortunately, a simple summation of the errors is not an adequate measure since opposing positive and negative errors can cancel one another. In fact, if we define the median values X and Y as

$$\bar{X} = \frac{1}{N} \sum_{i=1}^N x_i$$

Equation 3

and

$$\bar{Y} = \frac{1}{N} \sum_{i=1}^N y_i$$

Equation 4

then any line going through the point (X, Y) has a zero total error, irrespective of the values of a and b . The measure of the error that we normally use to avoid this problem is to use the sum of the squares of the errors as our measure of goodness of fit. This is written as

$$E_{total}^2 = \sum_{i=1}^N \varepsilon_i^2 = \sum_{i=1}^N [y_i - (a + bx_i)]^2$$

Equation 5

To minimize this value with respect to the choice of a and b , we equate the derivatives of Equation 5 (with respect to a, b) to zero and solve them simultaneously:

$$\frac{\partial \sum \varepsilon_i^2}{\partial a} = 0$$

$$\frac{\partial \sum \varepsilon_i^2}{\partial b} = 0$$

Equation 6

It can be shown that solving them simultaneously yields:

$$b = \frac{\sum x_i y_i - \frac{1}{N} \sum x_i \sum y_i}{\sum x_i^2 - \frac{1}{N} (\sum x_i)^2}$$

Equation 7

and

$$a = \bar{Y} - b\bar{X}$$

Equation 8

The above formulas represent the best values of a and b to minimize the sum of the square errors for the set of data chosen. This approach is called the *Method of Least Squares*.

Linear Example

Consider the data shown in the left three columns of Table 2 below.

Table 2. Data and analysis for a linear curve fit.

i	x	y	a+bx	(y-yb ar)^2	xy	x^2	y^2
1	26	92	93.99	548.9	2392	676	8464
2	30	85	89.85	269.9	2550	900	7225
3	44	78	75.37	88.9	3432	1936	6084
4	50	80	69.16	130.6	4000	2500	6400
5	62	54	56.75	212.3	3348	3844	2916
6	68	51	50.54	308.8	3468	4624	2601
7	74	40	44.34	816.3	2960	5476	1600
$\sum$	354	480		2375.7	22150	19956	35290
Mean Mean				r^2	a	b	
50.57 68.57				0.9249	120.881	-1.034	
				Syx	Sa	Sb	
				5.974	7.038	0.132	

Here, the standard deviations of the slope and intercept are high because of the spread of the errors. The correlation coefficient $r^2=0.925$, however, tells us that a linear approximation is a good fit to this data. The large variability in the data and the small number of data points have resulted in a large standard deviation of a y-estimate for a given value of x. The result of this analysis is depicted graphically in Figure 3.

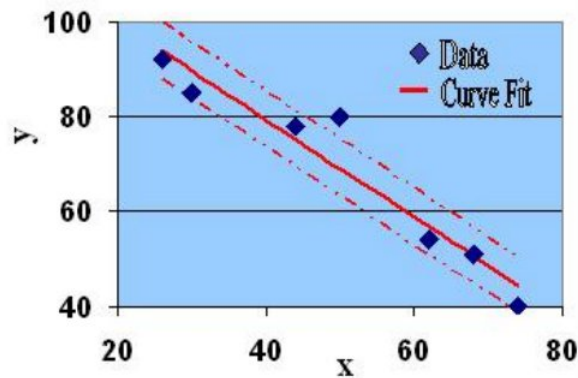


Figure 3. Linear curve fit showing lines at ± 1 standard deviation.

One characteristic inferred in Figure 3 is that the errors are not only “normal” at any given x , but the standard deviation is considered constant over all x . Thus, at any given x , we now know that the equation $y = mx + b$ predicts a value of $y = mx + b \pm S_{yx}$. In other words, if one took large amounts of data and plotted them on Figure 3, then 62.5% of all the data points would fall within the dotted lines at $\pm S_{yx}$ (above and below the line).

Alternate Curve Fits

It is important to note that the foregoing development is specific to a linear model for the data. The method shown, however, is general. Assume that we wish to represent the data with another (nonlinear) function, $y = f(x)$. The total square error is, therefore,

$$E^2 = \sum_{i=1}^N \varepsilon_i^2 = \sum_{i=1}^N [y_i - f(x_i)]^2$$

If there are m parameters in the function $f(x)$, then each derivative of the total squared error with respect to each parameter must be equated to zero and the system of equations solved for the parameters, λ_i .

$$\frac{\partial E^2}{\partial \lambda_i} = 0 \quad i=1..m$$

Note that we continue to use the Method of Least Squares, even though the function is not a linear one.

Exponential Curve Fit

Suppose we have data that, when plotted, appear to have an exponential character. If we choose an exponential function to represent the data, we write

$$y = ae^{bx}$$

In this formulation, a and b are the λ_i parameters we need to find that would best fit the function to the data. Thus, we write for the total squared error:

$$E^2 = \sum_{i=1}^N \varepsilon_i^2 = \sum_{i=1}^N [y_i - f(x_i)]^2 = \sum_{i=1}^N [y_i - ae^{bx_i}]^2$$

The two parameters are found by

$$\frac{\partial E^2}{\partial a} = 0 \quad \text{and} \quad \frac{\partial E^2}{\partial b} = 0 \quad \text{which are solved simultaneously to obtain}$$

$$b = \frac{\sum x_i \ln(y_i) - \frac{1}{N} (\sum x_i) (\sum \ln(y_i))}{(\sum x_i^2) - \frac{1}{N} (\sum x_i)^2}$$

$$a = \exp \left[\frac{1}{N} \sum \ln y_i - b \frac{\sum x_i}{n} \right]$$

It can be shown that this yields a Coefficient of Determination of

$$r^2 = \frac{\left[\sum x_i \ln y_i - \frac{1}{N} \sum x_i \sum \ln y_i \right]^2}{\left[\sum x_i^2 - \frac{(\sum x_i)^2}{N} \right] \left[\sum (\ln y_i)^2 - \frac{(\sum \ln y_i)^2}{N} \right]}$$

As in the linear case, a value of $r^2=1$ infers a "good fit" of the model to the data.

Exponential Example: Given the data in Table 3, find the appropriate exponential curve fit.

Table 3. Data for an exponential curve fit.

i	x	y
0	0.25	
1	0.72	2.16
2	1.31	1.61
3	1.95	1.16
4	2.58	0.85
5	3.14	0.50

The results of this analysis are shown in the figure below.

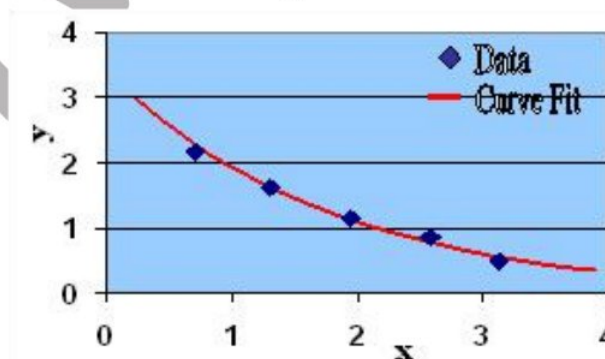


Figure 4. Results of an exponential curve fit.

Using the appropriate formulae for an exponential curve fit, we obtain $a = 3.45$, $b = -0.58$, and an correlation coefficient of $r^2 = 0.98$. As can be seen in the above graph, the function fits well as validated by the closeness of r^2 to 1.

Power Curve Fit

Suppose we have data that, when plotted, appear to have a power-law character. If we choose a power function to represent the data, we write

$$y = ax^b$$

As in the previous section, a and b are the λ_m parameters we seek that would best fit the function to the data. Thus, we write for the total squared error:

$$E^2 = \sum_{i=1}^N \varepsilon_i^2 = \sum_{i=1}^N [y_i - f(x_i)]^2 = \sum_{i=1}^N [y_i - ax_i^b]^2$$

The two parameters are found by

$$\frac{\partial E^2}{\partial a} = 0 \quad \text{and} \quad \frac{\partial E^2}{\partial b} = 0$$

which are solved simultaneously to obtain

$$b = \frac{\sum \ln x_i \ln(y_i) - \frac{1}{N} (\sum \ln x_i)(\sum \ln(y_i))}{\left(\sum (\ln x_i)^2 \right) - \frac{1}{N} (\sum \ln x_i)^2}$$

$$a = \exp \left[\frac{1}{N} \sum \ln y_i - b \frac{\sum \ln x_i}{N} \right]$$

It can be shown that this yields a Coefficient of Determination of

$$r^2 = \frac{\left[\sum \ln x_i \ln y_i - \frac{1}{N} \sum \ln x_i \sum \ln y_i \right]^2}{\left[\sum (\ln x_i)^2 - \frac{(\sum \ln x_i)^2}{N} \right] \left[\sum (\ln y_i)^2 - \frac{(\sum \ln y_i)^2}{N} \right]}$$

Power Law Example: Given the following data table (see the first three columns on the left), find the appropriate curve fit assuming a power law relationship between x and y . The spreadsheet shown as Table 4 also shows the added spreadsheet columns that help us compute a , b , and r^2 based on the formulae of Equations 25-27. The computed values of a , b , r^2 for a power curve fit are shown at the bottom of the spreadsheet.

Table 4. Data and analysis spreadsheet for a power law curve fit.

i	x	y	a x^b	ln(x)	ln(y)	ln(x)ln(y)	[ln(x)]^2	[ln(y)]^2
1	10	0.95	0.75	2.30	-0.05	-0.118	5.302	0.003
2	12	1.05	0.98	2.48	0.05	0.121	6.175	0.002
3	15	1.25	1.35	2.71	0.22	0.604	7.334	0.050
4	17	1.41	1.62	2.83	0.34	0.973	8.027	0.118
5	20	1.73	2.05	3.00	0.55	1.642	8.974	0.300
6	22	2.00	2.36	3.09	0.69	2.143	9.555	0.480
7	25	2.53	2.84	3.22	0.93	2.988	10.361	0.862
8	27	2.98	3.18	3.30	1.09	3.599	10.863	1.192
9	30	3.85	3.70	3.40	1.35	4.585	11.568	1.817
10	32	4.59	4.07	3.47	1.52	5.281	12.011	2.322
11	35	6.02	4.64	3.56	1.80	6.382	12.640	3.222
$\Sigma =$			245.0	28.4				
				33.35	8.49	28.201	102.810	10.369
				a	b	r^2		
				0.026	1.456	0.935538		

The graph below (Figure 5) shows the best fit curve to this data using the power law function. You will notice that the curve is low in the middle and high on each end.

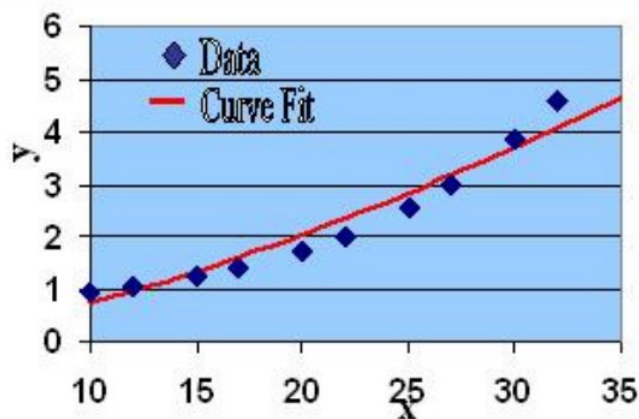


Figure 5. Results from a power law curve fit.

The disparity infers that a higher order power law fit may be more appropriate. The first step was to create a log-log plot of the data. A column was then created for $(y - y_0)$ (where $y_0 = \text{constant}$) and that data was included in the plot. The result was a linearized function was created in the log-log space by manually adjusting the value of y_0 . A model for a straight line with a slope of m on a log-log plot was used, or

$$\log(y - y_0) = \log(bx)^m = m \log(bx) = m \log(x) + m \log(b)$$

$$Y = mX + C \quad \text{where} \quad Y = \log(y - y_0), \quad X = \log(x)$$

I was able to fit this modified data much more accurately as depicted in Figure 6.

Problems on Curve Fitting:

Example 1: Find the least square line $y = a + bx$ for the data:

x	-2	-1	0	1	2
y	1	2	3	3	4

Solution:

x	y	x^2	xy
-2	1	4	-2
-1	2	1	-2
0	3	0	0
1	3	1	3
2	4	4	8
$\Sigma x=0$	$\Sigma y=13$	$\Sigma x^2=10$	$\Sigma xy=7$

$\therefore \Sigma x=0, \Sigma y=13, \Sigma x^2=10, \Sigma xy=7$, and $n=5$

The normal equations are

$$na + b\Sigma x = \Sigma y$$

$$a\Sigma x + b\Sigma x^2 = \Sigma xy$$

Putting the values of $n, \Sigma x, \Sigma y, \Sigma x^2, \Sigma xy$ in the above equation, we get

$$(12.3) 5a + b \cdot 0 = 13 \Rightarrow 5a = 13$$

$$(12.4) a \cdot 0 + b(10) = 7 \Rightarrow 10b = 7$$

From Eqs. (12.3) and (12.4) we get

$$a = 13/5 = 2.6, b = 7/10 = 0.7$$

The required of least square line is

$$y = 2.6 + (0.7)x$$

Example 2: Fit a straight line trend by the method of least square from the following data and find the trend values.

Year	1958	1959	1960	1961	1962
Sales (in lakhs of units)	65	95	80	115	105

Solution: We have $n=5$

$\therefore n$ is odd.

Taking middle year, i.e., 1960 as the origin. We get,

Year	Sales	x	x^2	xy
1958	65	-2	4	-130
1959	95	-1	1	-95
1960	80	0	0	0
1961	115	1	1	115
1962	105	2	4	210
Total	$\Sigma y=460$	$\Sigma x=0$	$\Sigma x^2=10$	$\Sigma xy=100$

$\therefore n=5, \Sigma x=0, \Sigma x^2=10, \Sigma y=460$, and $\Sigma xy=100$

$$a = \Sigma y/n \Rightarrow a = 460/5 = 92$$

$$b = \Sigma xy / \Sigma x^2 = 100/10 = 10$$

$\therefore$ The equation of the straight line trend is

$$y_c = a + bx \Rightarrow y_c = 92 + 10x$$

For the year 1958, $x = -2$

$$\Rightarrow y_c = y_{1958} = 92 + 10(-2) = 92 - 20 = 72$$

For the year 1959, $x = -1$

$$\Rightarrow y_c = y_{1959} = 92 + 10(-1) = 92 - 10 = 82$$

For the year 1960, $x = 0$

$$\Rightarrow y_c = y_{1960} = 92 + 10(0) = 92 + 0 = 92$$

For the year 1961, $x = 1$

$$\Rightarrow y_c = y_{1961} = 92 + 10(1) = 92 + 10 = 102$$

For the year 1962, $x = 2$

$$\Rightarrow y_c = y_{1962} = 92 + 10(2) = 92 + 20 = 112$$

We have

Year	Trend value
1958	72
1959	82
1960	92
1961	102
1962	112

and the straight line trend is $y_c = 92 + 10x$ or simply $y = 92 + 10x$.

Example 3: Determine the trend by the method of least squares. Also find the trend values.

Year	1950	1951	1952	1953	1954	1955	1956	1957
Value	346	411	392	512	626	640	611	796

Solution: Here $n = 9$

$\therefore n$ is even.

1953 and 1954 are the middle years.

The origin is $1953 + 1954 / 2 = 1953.5$

We take $x = \text{year} - 1953.5$

Year	x	y	x^2	xy
1950	-3.5	346	12.25	-1211.00
1951	-2.5	411	6.25	-1027.50
1952	-1.5	392	2.25	-588.00
1953	-0.5	512	0.25	-256.00
1954	0.5	626	0.25	313.00
1955	1.5	640	2.25	960.00
1956	2.5	611	6.25	1527.50
1957	3.5	796	12.25	2786.00
Total	0	4334	42.00	2504.00

$\therefore$ We have $n=8$, $\sum x=0$, $\sum y=4334$, $\sum x^2=42$, and $\sum xy=2504$

$$a = \frac{\sum y}{n} \Rightarrow a = \frac{4334}{8} = 541.75$$

$$b = \frac{\sum xy}{\sum x^2} = \frac{2504}{42} = 59.60$$

$\therefore$ The equation of the trend line is

$$y = 541.75 + (59.60)x$$

The trend values are:

Year	x	Trend: $y = 541.75 + (59.60)x$
1950	-3.5	333.15
1951	-2.5	392.75
1952	-1.5	452.35
1953	-0.5	511.95
1954	0.5	571.95
1955	1.5	631.15
1956	2.5	690.75
1957	3.5	750.35

UNIT-2 (ADVANCE ENGINEERING MATHEMATICS)

CORRELATION: CORRELATION COEFFICIENTS

LECTURE NO: -13

Correlation: Correlation Coefficients:

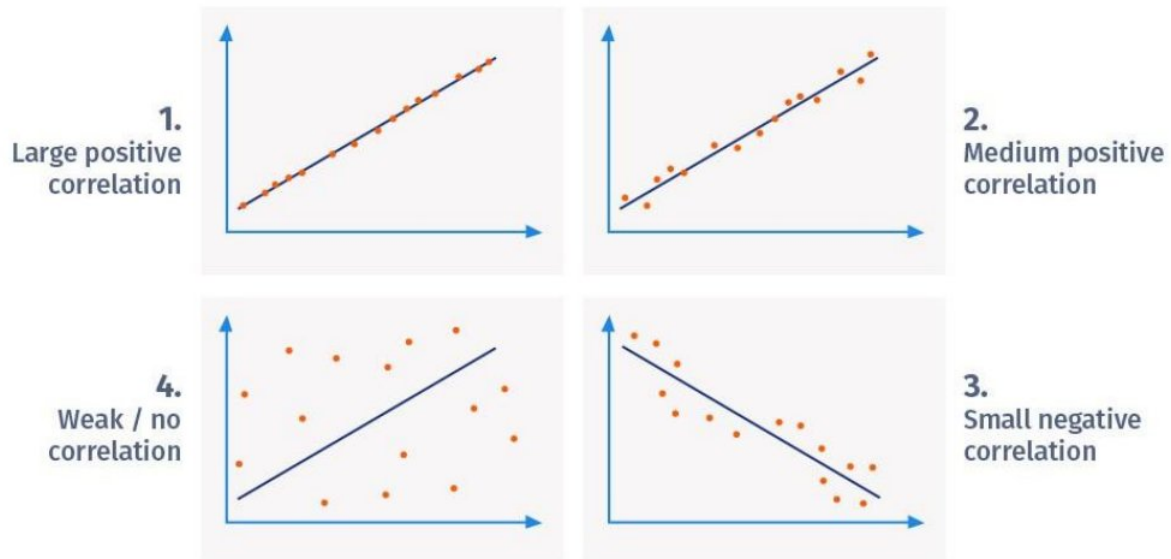
What is correlation analysis?

Correlation analysis in research is a statistical method used to measure the strength of the linear relationship between two variables and compute their association. Simply put - correlation analysis calculates the level of change in one variable due to the change in the other. A high correlation points to a strong relationship between the two variables, while a low correlation means that the variables are weakly related. When it comes to market research, researchers use correlation analysis to analyze quantitative data collected through research methods like surveys and live polls. They try to identify the relationship, patterns, significant connections, and trends between two variables or datasets. There is a positive correlation between two variables when an increase in one variable leads to the increase in the other. On the other hand, a negative correlation means that when one variable increases, the other decreases and vice-versa.

Example of correlation analysis

Correlation between two variables can be either a positive correlation, a negative correlation, or no correlation. Let's look at examples of each of these three types:

- 
- **Positive correlation:** A positive correlation between two variables means both the variables move in the same direction. An increase in one variable leads to an increase in the other variable and vice versa. For example, spending more time on a treadmill burns more calories.
 - **Negative correlation:** A negative correlation between two variables means that the variables move in opposite directions. An increase in one variable leads to a decrease in the other variable and vice versa. For example, increasing the speed of a vehicle decreases the time you take to reach your destination.
 - **Weak/Zero correlation:** No correlation exists when one variable does not affect the other. For example, there is no correlation between the number of years of school a person has attended and the letters in his/her name.



Uses of correlation analysis

Correlation analysis is used to study practical cases. Here, the researcher can't manipulate individual variables. For example, correlation analysis is used to measure the correlation between the patient's blood pressure and the medication used. Marketers use it to measure the effectiveness of advertising. Researchers measure the increase/decrease in sales due to a specific marketing campaign.

Advantages of correlation analysis

The advantages of correlation analysis are:

- Observe relationships: A correlation helps to identify the absence or presence of a relationship between two variables. It tends to be more relevant to everyday life.
- A good starting point for research: It proves to be a good starting point when a researcher starts investigating relationships for the first time.
- Uses for further studies: Researchers can identify the direction and strength of the relationship between two variables and later narrow the findings down in later studies.
- Simple metrics: Research findings are simple to classify. The findings can range from -1.00 to 1.00. There can be only three potential broad outcomes of the analysis.
- Correlation coefficient
- The degree of association is measured by a correlation coefficient, denoted by r . It is sometimes called Pearson's correlation coefficient after its originator and is a measure of linear association. If a curved line is needed to express the relationship, other and more complicated measures of the correlation must be used.
- The correlation coefficient is measured on a scale that varies from +1 through 0 to -1. Complete correlation between two variables is expressed by either +1 or -1. When one variable increases as the other increases the correlation is positive; when one decreases as the other increases it is negative. Complete absence of correlation is represented by 0
- Looking at data: scatter diagrams

- When an investigator has collected two series of observations and wishes to see whether there is a relationship between them, he or she should first construct a scatter diagram. The vertical scale represents one set of measurements and the horizontal scale the other. If one set of observations consists of experimental results and the other consists of a time scale or observed classification of some kind, it is usual to put the experimental results on the vertical axis. These represent what is called the “dependent variable”. The “independent variable”, such as time or height or some other observed classification, is measured along the horizontal axis, or baseline.
- The words “independent” and “dependent” could puzzle the beginner because it is sometimes not clear what is dependent on what. This confusion is a triumph of common sense over misleading terminology, because often each variable is dependent on some third variable, which may or may not be mentioned. It is reasonable, for instance, to think of the height of children as dependent on age rather than the converse but consider a positive correlation between mean tar yield and nicotine yield of certain brands of cigarette.’ The nicotine liberated is unlikely to have its origin in the tar: both vary in parallel with some other factor or factors in the composition of the cigarettes. The yield of the one does not seem to be “dependent” on the other in the sense that, on average, the height of a child depends on his age. In such cases it often does not matter which scale is put on which axis of the scatter diagram. However, if the intention is to make inferences about one variable from the other, the observations from which the inferences are to be made are usually put on the baseline. As a further example, a plot of monthly deaths from heart disease against monthly sales of ice cream would show a negative association. However, it is hardly likely that eating ice cream protects from heart disease! It is simply that the mortality rate from heart disease is inversely related – and ice cream consumption positively related – to a third factor, namely environmental temperature.

References:

Advance Engineering Mathematics, Chandrika Prasad, Standard Publishers and Distributors

UNIT-2 (ADVANCE ENGINEERING MATHEMATICS)

KARL CORRELATION COEFFICIENT

LECTURE NO: -14

Karl Correlation coefficient

Coefficient of Correlation

A coefficient of correlation is generally applied in statistics to calculate a relationship between two variables. The correlation shows a specific value of the degree of a linear relationship between the X and Y variables, say X and Y. There are various types of correlation coefficients. However, Pearson's correlation (also known as Pearson's R) is the correlation coefficient that is frequently used in linear regression.

Pearson's Coefficient Correlation

Karl Pearson's coefficient of correlation is an extensively used mathematical method in which the numerical representation is applied to measure the level of relation between linearly related variables. The coefficient of correlation is expressed by "r".

Karl Pearson Correlation Coefficient

$$r = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum(X - \bar{X})^2} \sqrt{\sum(Y - \bar{Y})^2}}$$

Where, $\bar{X}$ = mean of X variable

$\bar{Y}$ = mean of Y variable

Formula

Alternative Formula (covariance formula)

$$\text{Cov}(X, Y) = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{N} = \frac{\sum xy}{N}$$

Pearson correlation example

1. When a correlation coefficient is (1), that means for every increase in one variable, there is a positive increase in the other fixed proportion. For example, shoe sizes change according to the length of the feet and are perfect (almost) correlations.
2. When a correlation coefficient is (-1), that means for every positive increase in one variable, there is a negative decrease in the other fixed proportion. For example, the decrease in the quantity of gas in a gas tank shows a perfect (almost) inverse correlation with speed.
3. When a correlation coefficient is (0) for every increase, that means there is no positive or negative increase, and the two variables are not related.

Question: An experiment conducted on 9 different cigarette smoking subjects resulted in the following data –

Subject Number	Cigarettes smoked per week (averaged over the last 5 years of their life)	Number of years lived
1	25	63
2	35	68
3	10	72
4	40	62
5	85	65
6	75	46
7	60	51
8	45	60

9

50

55

Calculate the **correlation** of coefficient between the number of cigarettes smoked and the longevity of a test subject.

Solution

Let us first assign random variables to our data in the following way –

x – the number of cigarettes smoked

y – years lived

We'll be using the single formula for discrete data points here –

$$r_{xy} = \frac{N \sum x_i y_i - \sum x_i \sum y_i}{\sqrt{N \sum x_i^2 - (\sum x_i)^2} \sqrt{N \sum y_i^2 - (\sum y_i)^2}}$$

Let us now construct a table to compute all the values we are going to use in our correlation formula. Note that N here = 9

x	x ²	y	y ²	xy
25	625	63	3969	1575
35	1225	68	4624	2380
10	100	72	5184	720
40	1600	62	3844	2480

85	7225	65	4225	5525
75	5625	46	2116	3450
60	3600	51	2601	3060
45	2025	60	3600	2700
50	2500	55	3136	2750
$\Sigma x_i = 425$	$\Sigma x_i^2 = 24525$	$\Sigma y_i = 542$	$\Sigma y_i^2 = 33188$	$\Sigma x_i y_i = 24640$

$$(\Sigma x_i)^2 = 425^2 = 180625$$

$$(\Sigma y_i)^2 = 542^2 = 293764$$

using the values in the formula, we get –

$$\begin{aligned}
 r_{xy} &= \frac{N \Sigma x_i y_i - \Sigma x_i \Sigma y_i}{\sqrt{N \Sigma x_i^2 - (\Sigma x_i)^2}} \times \frac{\sqrt{N \Sigma y_i^2 - (\Sigma y_i)^2}}{N} \\
 &= \frac{9 \cdot 24640 - 425 \cdot 542}{\sqrt{9 \cdot 24525 - 180625}} \times \frac{\sqrt{9 \cdot 33188 - 293764}}{9} \\
 &= \frac{-8590}{40100} \times \frac{4928}{9} \\
 &= -0.61
 \end{aligned}$$

This implies a negative correlation between the considered variables i.e. The higher the number of cigarettes smoked per week in last 5 years, the lesser the number of years lived. Note that it DOES NOT mean that smoking cigarettes decreases the life span. Because, many other factors might be responsible for one's death. Still, it is an important conclusion nevertheless.

References:

Advance Engineering Mathematics, Chandrika Prasad, Standard Publishers and Distributors

UNIT-2 (ADVANCE ENGINEERING MATHEMATICS)

SPEARMAN RANK CORRELATION

LECTURE NO: - 15

Spearman rank correlation:

A plot of the data may reveal outlying points well away from the main body of the data, which could unduly influence the calculation of the correlation coefficient. Alternatively the variables may be quantitative discrete such as a mole count, or ordered categorical such as a pain score. A non-parametric procedure, due to Spearman, is to replace the observations by their ranks in the calculation of the correlation coefficient.

This results in a simple formula for Spearman's rank correlation, Rho.

$$r_s = 1 - \frac{6\sum d^2}{n(n^2 - 1)}$$

where d is the difference in the ranks of the two variables for a given individual.

Table 11.2 Derivation of Spearman rank correlation from data of table 11.1

Child number	Rank height	Rank dead space	d	d ²
1	1	3	2	4
2	2	1	-1	1
3	3	2	-1	1
4	4	4	0	0
5	5	5.5	0.5	0.25
6	6	11	5	25
7	7	7	0	0
8	8	5.5	-2.5	6.25
9	9	8	-1	1
10	10	13	3	9
11	11	10	-1	1
12	12	9	-3	9
13	13	12	-1	1
14	14	15	1	1
15	15	14	-1	1
Total				60.5

From this we get that

From this we get that

$$r_s = 1 - \frac{6 \times 60.5}{15 \times (225 - 1)} = (0.8920)$$

In this case the value is very close to that of the Pearson correlation coefficient. For $n > 10$, the Spearman rank correlation coefficient can be tested for significance using the t test given earlier.

Spearman's Rank correlation coefficient

A correlation can easily be drawn as a scatter graph, but the most precise way to compare several **pairs of data** is to use a statistical test - this establishes whether the correlation is really significant or if it could have been the result of chance alone.

Spearman's Rank correlation coefficient is a technique which can be used to summarise the strength and direction (negative or positive) of a relationship between two variables.

The result will always be between 1 and minus 1.

Method - calculating the coefficient

- Create a table from your data.
- Rank the two data sets. Ranking is achieved by giving the ranking '1' to the biggest number in a column, '2' to the second biggest value and so on. The smallest value in the column will get the lowest ranking. This should be done for both sets of measurements.
- Tied scores are given the mean (average) rank. For example, the three tied scores of 1 euro in the example below are ranked fifth in order of price, but occupy three positions (fifth, sixth and seventh) in a ranking hierarchy of ten. The mean rank in this case is calculated as $(5+6+7) \div 3 = 6$.
- Find the difference in the ranks (d): This is the difference between the ranks of the two values on each row of the table. The rank of the second value (price) is subtracted from the rank of the first (distance from the museum).
- Square the differences (d^2) To remove negative values and then sum them ($\sum d^2$).

Convenience Store	Distance from CAM (m)	Rank distance	Price of 50cl bottle (€)	Rank price	Difference between ranks (d)	d^2
1	50	10	1.80	2	8	64
2	175	9	1.20	3.5	5.5	30.25
3	270	8	2.00	1	7	49
4	375	7	1.00	6	1	1
5	425	6	1.00	6	0	0
6	580	5	1.20	3.5	1.5	2.25
7	710	4	0.80	9	-5	25
8	790	3	0.60	10	-7	49
9	890	2	1.00	6	-4	16
10	980	1	0.85	8	-7	49
						$\sum d^2 = 285.5$

Data Table: Spearman's Rank Correlation

- Calculate the coefficient (R_s) using the formula below. The answer will always be between 1.0 (a perfect positive correlation) and -1.0 (a perfect negative correlation).

When written in mathematical notation the Spearman Rank formula looks like this :

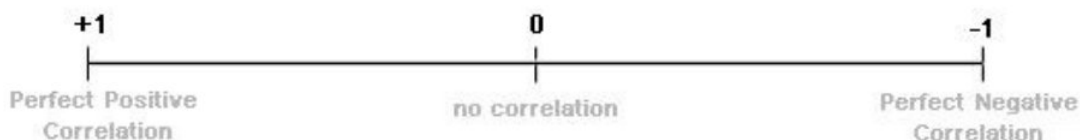
$$R_s = 1 - \left(\frac{6 \sum d^2}{n^3 - n} \right)$$

Now to put all these values into the formula.

- Find the value of all the d^2 values by adding up all the values in the Difference² column. In our example this is **285.5**. Multiplying this by **6** gives **1713**.
- Now for the bottom line of the equation. The value n is the number of sites at which you took measurements. This, in our example is **10**. Substituting these values into $n^3 - n$ we get **1000 - 10**
- We now have the formula: $R_s = 1 - (1713/990)$ which gives a value for R_s :
1 - 1.73 = -0.73

What does this R_s value of -0.73 mean?

The closer R_s is to +1 or -1, the stronger the likely correlation. A perfect positive correlation is +1 and a perfect negative correlation is -1. The R_s value of -0.73 suggests a fairly strong negative relationship.



A further technique is now required to test the **significance** of the relationship.

The R_s value of **-0.73** must be looked up on the Spearman Rank significance table below as follows:

- Work out the 'degrees of freedom' you need to use. This is the number of pairs in your sample minus 2 ($n-2$). In the example it is 8 ($10 - 2$).
- Now plot your result on the table.
- If it is below the line marked 5%, then it is possible your result was the product of chance and you must reject the hypothesis.
- If it is above the 0.1% significance level, then we can be 99.9% confident the correlation has not occurred by chance.
- If it is above 1%, but below 0.1%, you can say you are 99% confident.
- If it is above 5%, but below 1%, you can say you are 95% confident (i.e. statistically there is a 5% likelihood the result occurred by chance).

In the example, the value 0.73 gives a significance level of slightly less than 5%. That means that the probability of the relationship you have found being a chance event is **about 5 in a 100**. You are 95% certain that your hypothesis is correct. The reliability of your sample can be stated in terms of how many researchers completing the same study as yours would obtain the same results: 95 out of 100.

Graph of significance levels for Spearman's Rank correlation coefficients using Student's t distribution

- The fact two variables correlate cannot prove anything - only further research can actually prove that one thing affects the other.
- Data reliability is related to the size of the sample. The more data you collect, the more reliable your result.

Example Question:

The scores for nine students in physics and math are as follows:

- Physics: 35, 23, 47, 17, 10, 43, 9, 6, 28
- Mathematics: 30, 33, 45, 23, 8, 49, 12, 4, 31

Compute the student's ranks in the two subjects and compute the Spearman rank correlation.

Step 1: Find the ranks for each individual subject. I used the **Excel rank function** to find the ranks. If you want to rank by hand, order the scores from greatest to smallest; assign the rank 1 to the highest score, 2 to the next highest and so on:

Physics	Rank	Math	Rank
35	3	30	5
23	5	33	3
47	1	45	2
17	6	23	6
10	7	8	8
43	2	49	1
9	8	12	7
6	9	4	9
28	4	31	4

Step 2: Add a third column, d, to your data. The d is the difference between ranks. For example, the first student's physics rank is 3 and math rank is 5, so the difference is 3 points. In a fourth column, square your d values.

Physics	Rank	Math	Rank	d	d squared
35	3	30	5	2	4
23	5	33	3	2	4
47	1	45	2	1	1
17	6	23	6	0	0
10	7	8	8	1	1
43	2	49	1	1	1
9	8	12	7	1	1
6	9	4	9	0	0
28	4	31	4	0	0

Step 4: Sum (add up) all of your d-squared values.

$4 + 4 + 1 + 0 + 1 + 1 + 1 + 0 + 0 = 12$. You'll need this for the formula (the $\sum d^2$ is just "the sum of d-squared values").

Step 5: Insert the values into the formula. These ranks are not tied, so use the first formula:

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

$$= 1 - (6 \cdot 12) / (9(81 - 1))$$

$$= 1 - 72 / 720$$

$$= 1 - 0.1$$

= 0.9

The Spearman Rank Correlation for this set of data is 0.9.

SPEARMAN'S RANK CORRELATION COEFFICIENT

If the data are in ordinal scale then Spearman's rank correlation coefficient is used. It is denoted by the Greek letter ρ (**rho**).

Spearman's correlation can be calculated for the subjectivity data also, like competition scores. The data can be ranked from low to high or high to low by assigning ranks.

Spearman's rank correlation coefficient is given by the formula

$$\rho = 1 - \frac{6 \sum_{i=1}^n D_i^2}{n(n^2 - 1)}$$

where $D_i = R_{1i} - R_{2i}$

R_{1i} = rank of i in the first set of data

R_{2i} = rank of i in the second set of data and

n = number of pairs of observations

Interpretation

Spearman's rank correlation coefficient is a statistical measure of the strength of a monotonic (increasing/decreasing) relationship between paired data. Its interpretation is similar to that of Pearson's. That is, the closer to the ± 1 means the stronger the monotonic relationship.

Example

Two referees in a flower beauty competition rank the 10 types of flowers as follows:

Referee A	1	6	5	10	3	2	4	9	7	8
Referee B	6	4	9	8	1	2	3	10	5	7

Use the rank correlation coefficient and find out what degree of agreement is between the referees.

Solution:

Rank by 1 st referee R_{1i}	Rank by 2 nd referee R_{2i}	$D_i = R_{1i} - R_{2i}$	D_i^2
1	6	-5	25
6	4	2	4
5	9	-4	16
10	8	2	4
3	1	2	4
2	2	0	0
4	3	1	1
9	10	-1	1
7	5	2	4
8	7	1	1
			$\sum_{i=1}^n D_i^2 = 60$

Here $n = 10$ and $\sum_{i=1}^n D_i^2 = 60$

$$\rho = 1 - \frac{6 \sum_{i=1}^n D_i^2}{n(n^2 - 1)}$$

$$= 1 - \frac{6 \times 60}{10(10^2 - 1)} = 1 - \frac{360}{10(99)} = 1 - \frac{360}{990} = 0.636$$

Interpretation: Degree of agreement between the referees 'A' and 'B' is 0.636 and they have "strong agreement" in evaluating the competitors.

Example

Quotations of index numbers of equity share prices of a certain joint stock company and the prices of preference shares are given below.

ence shares are given below.

Years	2013	2014	2015	2016	2017	2008	2009
Equity shares	97.5	99.4	98.6	96.2	95.1	98.4	97.1
Reference shares	75.1	75.9	77.1	78.2	79	74.6	76.2

Using the method of rank correlation determine the relationship between equity shares and preference shares prices.

Solution:

Equity shares	Preference share	R_{1i}	R_{2i}	$D_i = R_{1i} - R_{2i}$	D_i^2
97.5	75.1	4	6	-2	4
99.4	75.9	1	5	-4	16
98.6	77.1	2	3	-1	1
96.2	78.2	6	2	4	16
95.1	79.0	7	1	6	36
98.4	74.6	3	7	-4	16
97.1	76.2	5	4	1	1
					$\sum_{i=1}^n D_i^2 = 90$

$$\sum_{i=1}^n D_i^2 = 90 \text{ and } n = 7.$$

Rank correlation coefficient is

$$\begin{aligned} \rho &= 1 - \frac{6 \sum_{i=1}^n D_i^2}{n(n^2 - 1)} \\ &= 1 - \frac{6 \times 90}{7(7^2 - 1)} = 1 - \frac{540}{7 \times 48} = 1 - \frac{540}{336} = 1 - 1.6071 = -0.6071 \end{aligned}$$

Interpretation: There is a negative correlation between equity shares and preference share prices.

There is a strong disagreement between equity shares and preference share prices.

Repeated ranks

When two or more items have equal values (i.e., a tie) it is difficult to give ranks to them. In such cases the items are given the average of the ranks they would have received. For example, if two individuals are placed in the 8th place, they are given the rank $[8+9] / 2 = 8.5$ each, which is common rank to be assigned and the next will be 10; and if three ranked equal at the 8th place, they are given the rank $[8 + 9 + 10] / 3 = 9$ which is the common rank to be assigned to each; and the next rank will be 11.

In this case, a different formula is used when there is more than one item having the same value.

$$\rho = 1 - 6 \left[\frac{\sum D_i^2 + \frac{1}{12}(m_1^3 - m_1) + \frac{1}{12}(m_2^3 - m_2) + \dots}{n(n^2 - 1)} \right]$$

where m_i is the number of repetitions of i^{th} rank

Example

Compute the rank correlation coefficient for the following data of the marks obtained by 8 students in the Commerce and Mathematics.

Commerce and Mathematics.

Marks in Commerce	15	20	28	12	40	60	20	80
Marks in Mathematics	40	30	50	30	20	10	30	60

Solution:

Marks in Commerce (X)	Rank (R_{1i})	Marks in Mathematics (Y)	Rank (R_{2i})	$D_i = R_{1i} - R_{2i}$	D_i^2
15	2	40	6	-4	16
20	3.5	30	4	-0.5	0.25
28	5	50	7	-2	4
12	1	30	4	-3	9
40	6	20	2	4	16
60	7	10	1	6	36
20	3.5	30	4	-0.5	0.25
80	8	60	8	0	0
				Total	$\sum D^2 = 81.5$

$$\rho = 1 - 6 \left[\frac{\sum D_i^2 + \frac{1}{12}(m_1^3 - m_1) + \frac{1}{12}(m_2^3 - m_2) + \dots}{n(n^2 - 1)} \right]$$

Repetitions of ranks

In Commerce (X), 20 is repeated two times corresponding to ranks 3 and 4. Therefore, 3.5 is assigned for rank 2 and 3 with $m_1=2$.

In Mathematics (Y), 30 is repeated three times corresponding to ranks 3, 4 and 5. Therefore, 4 is assigned for ranks 3,4 and 5 with $m_2=3$.

Therefore,

$$\rho = 1 - 6 \left[\frac{81.5 + \frac{1}{12}(2^3 - 2) + \frac{1}{12}(3^3 - 3)}{8(8^2 - 1)} \right]$$
$$= 1 - 6 \frac{[81.5 + 0.5 + 2]}{504} = 1 - \frac{504}{504} = 0$$

References:

Advance Engineering Mathematics, Chandrika Prasad, Standard Publishers and Distributors

UNIT-2 (ADVANCE ENGINEERING MATHEMATICS)

LINEAR REGRESSION

LECTURE NO: -16

Linear Regression:

Linear regression attempts to model the relationship between two variables by fitting a linear equation to observed data. One variable is considered to be an explanatory variable, and the other is considered to be a dependent variable. For example, a modeler might want to relate the weights of individuals to their heights using a linear regression model.

Before attempting to fit a linear model to observed data, a modeler should first determine whether or not there is a relationship between the variables of interest. This does not necessarily imply that one variable *causes* the other (for example, higher SAT scores do not *cause* higher college grades), but that there is some significant association between the two variables. A scatterplot can be a helpful tool in determining the strength of the relationship between two variables. If there appears to be no association between the proposed explanatory and dependent variables (i.e., the scatterplot does not indicate any increasing or decreasing trends), then fitting a linear regression model to the data probably will not provide a useful model. A valuable numerical measure of association between two variables is the correlation coefficient, which is a value between -1 and 1 indicating the strength of the association of the observed data for the two variables.

A linear regression line has an equation of the form $Y = a + bX$, where X is the explanatory variable and Y is the dependent variable. The slope of the line is b , and a is the intercept (the value of y when $x = 0$).

Multiple Regression

Simple linear regression is a technique for predicting the value of a dependent variable, based on the value of a single independent variable. Sometimes, you only need one relevant independent variable to make an accurate prediction.

More often, however, the prediction is better when you use two or more independent variables. Multiple regression is a technique for predicting the value of a dependent variable, based on the values of two or more independent variables.

The Regression Equation

This is a tutorial about *linear* regression, so our focus is on *linear* relationships between variables. The regression equation that expresses the linear relationships between a single dependent variable and one or more independent variables is:

$$\hat{y} = b_0 + b_1x_1 + b_2x_2 + \dots + b_{k-1}x_{k-1} + b_kx_k$$

In this equation, $\hat{y}$ is the *predicted* value of the dependent variable. Values of the k independent variables are denoted by $x_1, x_2, x_3, \dots, x_k$.

And finally, we have the b 's - $b_0, b_1, b_2, \dots, b_k$. The b 's are constants, called regression coefficients. Values are assigned to the b 's based on the principle of least squares.

What is the Principle of Least Squares?

In multiple regression, the deviation of the actual value for a dependent variable from its predicted value is called the residual. The residual (e) for a single observation i is:

$$e_i = y_i - \hat{y}_i = y_i - (b_0 + b_1x_{1i} + b_2x_{2i} + \dots + b_kx_{ki})$$

Assume that the set of data consists of n observations. The principle of least squares requires that the sum of squared residuals for all n observations be minimized. That is, we want the following value to be as small as possible:

$$\sum [y_i - (b_0 + b_1x_{1i} + b_2x_{2i} + \dots + b_kx_{ki})]^2$$

Regression analysis requires that the values of $b_0, b_1, \dots, b_k$ be defined to minimize the sum of the squared residuals. When we assign values to regression coefficients in this way, we are following the principle of least squares.

Normal Equations for Simple Regression

Finding the right values for regression coefficients (i.e., values that satisfy a least squares criterion) involves solving a set of linear equations. These equations can be derived using calculus, and they are called normal equations.

To illustrate the use of normal equations, let's look at simple linear regression - regression with one dependent variable (y) and one independent variable (x). With simple linear regression, the regression equation is:

$$\hat{y} = b_0 + b_1x$$

The normal equations for simple linear regression are:

$$\sum y_i = nb_0 + b_1(\sum x_i)$$

$$\sum x_i y_i = b_0(\sum x_i) + b_1(\sum x_i^2)$$

Here, we have two equations with two unknowns. The unknowns are the regression coefficients b_0 and b_1 . Using ordinary algebra, we can solve for b_0 and b_1 . The result is:

$$b_1 = \frac{\sum [(x_i - \bar{x})(y_i - \bar{y})]}{\sum [(x_i - \bar{x})^2]}$$

$$b_0 = \bar{y} - b_1 * \bar{x}$$

where $\bar{x}$ is the mean x score, and $\bar{y}$ is the mean y score. Note that these are the same equations that we presented in a previous lesson, when we introduced the topic of simple linear regression.

The use of normal equations to assign values to regression coefficients becomes more complicated when there are two or more independent variables. We'll tackle that challenge in the next lesson.

Regression is used in statistical modeling and it basically tells us the relationship between variables and their movement in the future. Apart from statistical methods like standard deviation, regression, correlation. The regression analysis is the most widely and commonly accepted measure to measure the variance in the industry. These relationships are seldom exact because there is variation caused by many variables, not just the variables being studied. The method is widely used in the industry for predictive modeling and forecasting measures. Regression tells us the relationship of the independent variable on the dependent variable and to explore the forms of these relationships.

The formula for Regression Analysis –

$$Y = a + bX + \epsilon$$

- **Y** = Stands for the dependent variable
- **X** = Stands for an independent variable
- **a** = Stands for the intercept
- **b** = Stands for the slope
- **ϵ** = Stands for the error term

The formula for intercept “a” and the slope “b” can be calculated as per below.

$$a = \frac{(\sum y)(\sum x^2) - (\sum x)(\sum xy)}{n(\sum x^2) - (\sum x)^2}$$

$$b = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$$

Regression analysis is one of the most powerful multivariate statistical technique as the user can interpret parameters the slope and the intercept of the functions that link with two or more variables in a given set of data.

There are two types of regression multilinear regression and simple linear regression. The simple linear regression is explained and is the same as $aY = a + bX_1 + cX_2 + dX_3 + \epsilon$

Where,

- Y – Dependent variable
- X_1, X_2, X_3 – Independent (explanatory) variables
- a – Intercept
- b, c, d – Slopes
- ϵ – Residual (error)

Example #1

Consider the following two variables x and y , you are required to do the calculation of the regression.

	A	B	C
1	X	Y	
2	34.86	43.04	
3	42.58	51.88	
4	71.73	88.55	
5	110.77	130.69	
6	259.95	314.17	
7			

Solution:

Using the above formula, we can do the calculation of [linear regression in excel](#) as follows.

	A	B	C	D	E
1	X	Y	XY	X ²	Y ²
2	34.86	43.04	1500.3744	1215.2196	1852.4416
3	42.58	51.88	2209.0504	1813.0564	2691.5344
4	71.73	88.55	6351.6915	5145.1929	7841.1025
5	110.77	130.69	14476.5313	12269.9929	17079.8761
6	259.95	314.17	81668.4915	67574.0025	98702.7889
7	(Σx) = 519.89	(Σy) = 628.33	(ΣXY) = 106206.14	(ΣX ²) = 88017.46	(ΣY ²) = 128167.74
8					

We have all the values in the above table with n = 5.

Now, first, calculate the intercept and slope for the regression.

Calculation of Intercept is as follows,

C9		f_x	$=((B7*D7)-(A7*C7))/(5*D7-A7^2)$		
1	A	B	C	D	E
1	X	Y	XY	X ²	Y ²
2	34.86	43.04	1500.3744	1215.2196	1852.4416
3	42.58	51.88	2209.0504	1813.0564	2691.5344
4	71.73	88.55	6351.6915	5145.1929	7841.1025
5	110.77	130.69	14476.5313	12269.9929	17079.8761
6	259.95	314.17	81668.4915	67574.0025	98702.7889
7	519.89	628.33	106206.14	88017.46	128167.74
8					
9	a (intercept)		0.52		
10					

$$a = \frac{(\Sigma y)(\Sigma x^2) - (\Sigma x)(\Sigma xy)}{n(\Sigma x^2) - (\Sigma x)^2}$$

$$a = (628.33 * 88,017.46) - (519.89 * 106,206.14) / 5 * 88,017.46 - (519.89)^2$$

$$a = 0.52$$

Calculation of Slope is as follows,

C9		f_x	$=((5*C7)-(A7*B7))/(5*D7-A7^2)$		
	A	B	C	D	
1	X	Y	XY	X ²	$b = n(\sum xy) - (\sum x)(\sum y) / n(\sum x^2) - (\sum x)^2$
2	34.86	43.04	1500.3744	1215.2196	1852.4416
3	42.58	51.88	2209.0504	1813.0564	2691.5344
4	71.73	88.55	6351.6915	5145.1929	7841.1025
5	110.77	130.69	14476.5313	12269.9929	17079.8761
6	259.95	314.17	81668.4915	67574.0025	98702.7889
7	519.89	628.33	106206.14	88017.46	128167.74
8					
9	b (slope)		1.20		
10					

$$b = (5 * 106,206.14) - (519.89 * 628.33) / (5 * 88,017.46) - (519,89)^2$$

$$b = 1.20$$

Let's now input the values in the regression formula to get regression.

Hence the regression line $Y = 0.52 + 1.20 * X$

References:

Advance Engineering Mathematics, Chandrika Prasad, Standard Publishers and Distributors

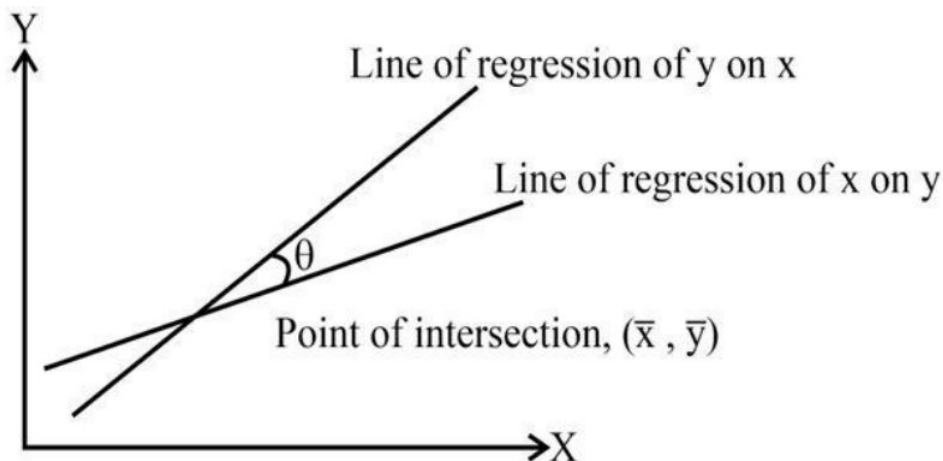
UNIT-2 (ADVANCE ENGINEERING MATHEMATICS)

Angle Between the Two Lines of Regression

LECTURE NO: -17

Angle Between the Two Lines of Regression

We have seen that the two lines of regression intersect at the point $(\bar{x}, \bar{y})$. Therefore, for a correlation that is not perfect, the two lines will be at an angle (θ) to each other.



The line of regression of x on y is given by the equation

$$(x - \bar{x}) = (r\sigma_x/\sigma_y)(y - \bar{y})$$

Let the slope of this line be m_1 .

We can rewrite the line equation as

$$(y - \bar{y}) = (\sigma_y r / \sigma_x)(x - \bar{x})$$

When we compare this with the standard line equation, $y = mx + c$, we get

$$m_1 = (\sigma_y r / \sigma_x)$$

Similarly, the line of regression of y on x is given by the equation

$$(y - \bar{y}) = r\sigma_y\sigma_x(x - \bar{x}) \quad (y - \bar{y}) = r\sigma_y\sigma_x(x - \bar{x})$$

Here, let the slope of this line be m_2 .

When we compare this with the standard line equation, $y = mx + c$, we get

$$m_2 = r\sigma_y\sigma_x$$

For two intersecting lines with slopes m_1 and m_2 , the angle between these two lines, θ , can be found by the formula,

$$\tan\theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \quad \tan\theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\Rightarrow \tan\theta = \left| \frac{\sigma_y r_{xy} - r_{yx} \sigma_x}{\sigma_y \sigma_x (1 + r_{xy} r_{yx})} \right| \Rightarrow \tan\theta = \left| \frac{\sigma_y r_{xy} - r_{yx} \sigma_x}{\sigma_y \sigma_x (1 + r_{xy} r_{yx})} \right|$$

$$\Rightarrow \tan\theta = \left| \frac{\sigma_y \sigma_x (1 - r)}{\sigma_y \sigma_x (1 + r)} \right| \Rightarrow \tan\theta = \left| \frac{\sigma_y \sigma_x (1 - r)}{\sigma_y \sigma_x (1 + r)} \right|$$

$$\Rightarrow \tan\theta = \left| \frac{\sigma_y \sigma_x (1 - r^2)}{\sigma_x^2 \sigma_y^2 + \sigma_y^2} \right| \Rightarrow \tan\theta = \left| \frac{\sigma_y \sigma_x (1 - r^2)}{\sigma_x^2 \sigma_y^2 + \sigma_y^2} \right|$$

$$\therefore \tan\theta = \left| \frac{(1 - r^2) \sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \right| \therefore \tan\theta = \left| \frac{(1 - r^2) \sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \right|$$

$$\Rightarrow \theta = \tan^{-1} \left| \frac{(1 - r^2) \sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \right| \Rightarrow \theta = \tan^{-1} \left| \frac{(1 - r^2) \sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \right|$$

PRACTICE EXAM QUESTIONS AT EMBIBE

Type of Correlation and Angle of Regression

Following are the different types of Correlation and Angle of Regression

1. If we substitute $r=0$ in the equation of θ we get,

$$\theta = \tan^{-1} \left| \frac{(1 - 0^2) \sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \right| \Rightarrow \theta = \tan^{-1} \left| \frac{(1 - 0^2) \sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \right|$$

$$\Rightarrow \theta = \tan^{-1} \Rightarrow \theta = \tan^{-1} (\text{not defined})$$

$$\therefore \theta = \frac{\pi}{2} \therefore \theta = \frac{\pi}{2}$$

It means that when $r=0$, the angle between the lines of regression is $\theta = \frac{\pi}{2}$.

$r=0$ means the variables x and y have no correlation.

Thus, for uncorrelated variables x and y , the lines of regression are perpendicular to each other.

2. If we substitute $r=\pm 1$ in the equation of θ we get,

$$\theta = \tan^{-1} \left| \frac{[1 - (\pm 1)^2] \sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \right| \Rightarrow \theta = \tan^{-1} \left| \frac{[1 - (\pm 1)^2] \sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \right|$$

$$\Rightarrow \theta = \tan^{-1}(0) \Rightarrow \theta = \tan^{-1}(0)$$

$$\therefore \theta = 0 \therefore \theta = 0 \text{ or } \theta = \pi$$

It means that when $r=\pm 1$, the angle between the lines of regression is $\theta = 0$ or $\theta = \pi$.

$r=\pm 1$ means the variables have a perfect positive or perfect negative correlation.

Thus, for perfectly correlated (positive or negative) variables x and y , the lines of regression are coincident.

3. The modulus $(| |)$ in the formula of θ indicates that it can take two values.

For a single value of $r > 0$,

- $\tan^{-1}\frac{(1-r^2)\sigma_x\sigma_y}{\sigma_x^2+\sigma_y^2}$ will give the **acute angle** between the two lines of regression, i.e., $0 < \theta < \frac{\pi}{2}$.
- $\tan^{-1}\frac{(r^2-1)\sigma_x\sigma_y}{\sigma_x^2+\sigma_y^2}$ will give the **obtuse angle** between the two lines of regression, i.e., $\frac{\pi}{2} < \theta < \pi$.

4. As the angle between the two lines of regression decreases from $\frac{\pi}{2}$ to 0, the correlation between the variables increases from 0 to 1.

5. As the angle between the two lines of regression increases from $\frac{\pi}{2}$ to π , the correlation between the variables increases from 0 to -1.

Solved Examples – Angle Between Two Lines of Regression

Below are a few solved examples that can help in getting a better idea.

Q.1. For two variables, x and y , the correlation coefficient is 0.5. The acute angle between the two lines of regression is $\tan^{-1}35$. Show that $\sigma_x = 12\sigma_y$.

Ans: Correlation coefficient, $r = 0.5$

Angle between the two lines of regression, $\theta = \tan^{-1}35$

Substituting these values in the equation of θ ,

$$\theta = \tan^{-1} \left| \frac{(1-r^2)\sigma_x\sigma_y}{\sigma_x^2+\sigma_y^2} \right| \quad \theta = \tan^{-1} \left| \frac{(1-0.5^2)\sigma_x\sigma_y}{\sigma_x^2+\sigma_y^2} \right|$$

Since, the angle given is acute, we can write

$$\theta = \tan^{-1}\frac{(1-r^2)\sigma_x\sigma_y}{\sigma_x^2+\sigma_y^2} \quad \theta = \tan^{-1}\frac{(1-0.5^2)\sigma_x\sigma_y}{\sigma_x^2+\sigma_y^2}$$

$$\therefore \tan^{-1}35 = \tan^{-1}\frac{(1-0.5^2)\sigma_x\sigma_y}{\sigma_x^2+\sigma_y^2} \quad \therefore \tan^{-1}35 = \tan^{-1}\frac{(1-0.25)\sigma_x\sigma_y}{\sigma_x^2+\sigma_y^2}$$

$$\Rightarrow 35 = \frac{(1-0.25)\sigma_x\sigma_y}{\sigma_x^2+\sigma_y^2} \Rightarrow 35 = \frac{0.75\sigma_x\sigma_y}{\sigma_x^2+\sigma_y^2}$$

$$\Rightarrow 35(\sigma_x^2+\sigma_y^2) = 0.75\sigma_x\sigma_y \Rightarrow 35\sigma_x^2 + 35\sigma_y^2 = 0.75\sigma_x\sigma_y$$

$$\Rightarrow 35\sigma_x^2 - 0.75\sigma_x\sigma_y + 35\sigma_y^2 = 0$$

$$\Rightarrow 35(\sigma_x^2 + \sigma_y^2) = 0.75\sigma_x\sigma_y \Rightarrow 35(\sigma_x^2 + \sigma_y^2) = 0.75\sigma_x\sigma_y$$

$$\Rightarrow 2(\sigma_x^2 + \sigma_y^2) = 5(\sigma_x\sigma_y) \Rightarrow 2(\sigma_x^2 + \sigma_y^2) = 5(\sigma_x\sigma_y)$$

$$\Rightarrow 2\sigma_x^2 - 5\sigma_x\sigma_y + 2\sigma_y^2 = 0 \Rightarrow 2\sigma_x^2 - 5\sigma_x\sigma_y + 2\sigma_y^2 = 0$$

$$\Rightarrow 2\sigma_x^2 - 4\sigma_x\sigma_y - \sigma_x\sigma_y + 2\sigma_y^2 = 0 \Rightarrow 2\sigma_x^2 - 4\sigma_x\sigma_y - \sigma_x\sigma_y + 2\sigma_y^2 = 0$$

$$\Rightarrow 2\sigma_x(\sigma_x - 2\sigma_y) - \sigma_y(\sigma_x - 2\sigma_y) = 0 \Rightarrow 2\sigma_x(\sigma_x - 2\sigma_y) - \sigma_y(\sigma_x - 2\sigma_y) = 0$$

$$\Rightarrow (\sigma_x - 2\sigma_y)(2\sigma_x - \sigma_y) = 0 \Rightarrow (\sigma_x - 2\sigma_y)(2\sigma_x - \sigma_y) = 0$$

$$\Rightarrow (\sigma_x - 2\sigma_y) = 0 \Rightarrow (\sigma_x - 2\sigma_y) = 0 \text{ or } (2\sigma_x - \sigma_y) = 0$$

$$\Rightarrow (2\sigma_x - \sigma_y) = 0 \Rightarrow (2\sigma_x - \sigma_y) = 0$$

$$\therefore \sigma_x = 12\sigma_y$$

Q.2. For two variables, x and y , regression coefficients are $b_{xy}=0.4$ and $b_{yx}=1.6$. If θ is the angle between the two regression lines, then find the value of θ .

Ans: We know that,

$$\tan \theta = \left| \frac{(1-r^2)\sigma_x\sigma_y}{\sigma_x^2 + \sigma_y^2} \right| \quad \tan \theta = \left| \frac{(1-r^2)\sigma_x\sigma_y}{\sigma_x^2 + \sigma_y^2} \right|$$

$$\Rightarrow \tan \theta = \left| \frac{(1-r^2)\sigma_x\sigma_y}{\sigma_x^2 + \sigma_y^2} \right| \Rightarrow \tan \theta = \left| \frac{(1-r^2)\sigma_x\sigma_y}{\sigma_x^2 + \sigma_y^2} \right|$$

$$\Rightarrow \tan \theta = \left| \frac{(1-r^2)\sigma_x\sigma_y}{\sigma_x^2 + \sigma_y^2} \right| \Rightarrow \tan \theta = \left| \frac{(1-r^2)\sigma_x\sigma_y}{\sigma_x^2 + \sigma_y^2} \right|$$

$$\Rightarrow \tan \theta = \left| \frac{(1-r^2)\sigma_x\sigma_y}{\sigma_x^2 + \sigma_y^2} \right| \Rightarrow \tan \theta = \left| \frac{(1-r^2)\sigma_x\sigma_y}{\sigma_x^2 + \sigma_y^2} \right|$$

$$\therefore \tan \theta = \left| \frac{1-r^2}{1-r^2} \right| \therefore \tan \theta = \left| \frac{1-r^2}{1-r^2} \right|$$

Now,

$$r\sigma_x\sigma_y = b_{xy}\sigma_x\sigma_y = b_{xy}$$

$$r\sigma_y\sigma_x = b_{yx}\sigma_y\sigma_x = b_{yx}$$

$$r^2 = b_{xy}b_{yx} \quad r^2 = b_{xy}b_{yx}$$

$$\therefore \tan \theta = \left| \frac{1-b_{xy}b_{yx}}{1-b_{xy}b_{yx}} \right| \therefore \tan \theta = \left| \frac{1-b_{xy}b_{yx}}{1-b_{xy}b_{yx}} \right|$$

This is one more formula for the angle between two lines of regression.

$$\therefore \tan \theta = \left| \frac{1-0.4 \times 1.6}{1-0.4 \times 1.6} \right| \therefore \tan \theta = \left| \frac{1-0.64}{1-0.64} \right|$$

$$\Rightarrow \tan \theta = \left| \frac{1-0.64}{1-0.64} \right| \Rightarrow \tan \theta = \left| \frac{1-0.64}{1-0.64} \right|$$

$$\Rightarrow \tan \theta = \left| \frac{0.36}{0.36} \right| \Rightarrow \tan \theta = \left| \frac{0.36}{0.36} \right|$$

$$\therefore \tan \theta = 0.18 \therefore \tan \theta = 0.18$$

Q.3. If the standard deviation of y is twice the standard deviation of x , find the tangent of the acute angle between the two lines of regression if the correlation coefficient is 0.25.

Ans: Standard deviation of $y = 2 \times$ (Standard deviation of x) $\sigma_y = 2\sigma_x$

$$\therefore \sigma_y = 2\sigma_x \therefore \sigma_y = 2\sigma_x$$

Correlation coefficient, $r = 0.25$

We know that for the acute angle between the two lines of regression,

$$\tan \theta = \frac{(1-r^2)\sigma_x\sigma_y}{\sigma_x^2 + \sigma_y^2} \quad \tan \theta = \frac{(1-r^2)\sigma_x\sigma_y}{\sigma_x^2 + \sigma_y^2}$$

$$\Rightarrow \tan \theta = \frac{(1-0.25^2)\sigma_x(2\sigma_x)}{\sigma_x^2 + (2\sigma_x)^2} \Rightarrow \tan \theta = \frac{(1-0.0625)\sigma_x(2\sigma_x)}{\sigma_x^2 + 4\sigma_x^2}$$

$$\Rightarrow \tan \theta = \frac{(1-0.0625)2\sigma_x^2}{5\sigma_x^2} \Rightarrow \tan \theta = \frac{(1-0.0625)2}{5}$$

$$\Rightarrow \tan \theta = 0.93750.25 \times 25 \Rightarrow \tan \theta = 0.93750.25 \times 25$$

$$\therefore \tan \theta = 1.5 \therefore \tan \theta = 1.5$$

Q.4. The regression lines of xx on yy and yy on xx are $5x - y = 6$ and $2x - 3y = -8$, respectively. Find the angle between these lines.

Ans: **Line of regression of xx on yy**

$$5x - y = 6$$

$$\Rightarrow 5x = y + 6 \Rightarrow x = \frac{y+6}{5}$$

$$\Rightarrow x = \frac{y}{5} + \frac{6}{5} \Rightarrow x = 0.2y + 1.2$$

$$\Rightarrow x = 0.2y + 1.2 \Rightarrow x = 0.2y + 1.2$$

$$\therefore b_{xy} = 0.2 \therefore b_{xy} = 0.2$$

Line of regression of yy on xx

$$2x - 3y = -8$$

$$\Rightarrow -3y = -2x - 8 \Rightarrow -3y = -2x - 8$$

$$\Rightarrow y = \frac{-2x - 8}{-3} \Rightarrow y = \frac{2x}{3} + \frac{8}{3}$$

$$\Rightarrow y = \frac{2}{3}x + \frac{8}{3} \Rightarrow y = 0.67x + 2.67$$

$$\therefore b_{yx} = \frac{2}{3} \therefore b_{yx} = 0.67$$

Angle between the two lines of regression,

$$\theta = \tan^{-1} \left| \frac{1 - (0.2)(0.67)}{0.2 + 0.67} \right| \Rightarrow \theta = \tan^{-1} \left| \frac{1 - 0.134}{0.87} \right|$$

$$\Rightarrow \theta = \tan^{-1} \left| \frac{0.866}{0.87} \right| \Rightarrow \theta = \tan^{-1} \left| \frac{0.866}{0.87} \right|$$

$$\Rightarrow \theta = \tan^{-1} \left| \frac{0.866}{0.87} \right| \Rightarrow \theta = \tan^{-1} \left| \frac{0.866}{0.87} \right|$$

$$\Rightarrow \theta = \tan^{-1} \left| \frac{0.866}{0.87} \right| \Rightarrow \theta = \tan^{-1} \left| \frac{0.866}{0.87} \right|$$

$$\Rightarrow \theta = \tan^{-1} \left| \frac{0.866}{0.87} \right| \Rightarrow \theta = \tan^{-1} \left| \frac{0.866}{0.87} \right|$$

$$\therefore \theta = \frac{\pi}{4} \therefore \theta = \frac{\pi}{4}$$

This is the acute angle between the two lines of regression.

Q.5. The two lines of regression are $6x + 15y = 27$ and $6x + 3y = 15$. Calculate the angle between the two lines of regression.

Ans: We do not know which of these lines is the line of regression of xx on yy and the line of regression of yy on xx.

Assume that the line of regression of xx on yy is $6x + 15y = 27$, and the line of regression of yy on xx is $6x + 3y = 15$.

We now represent these equations in proper form.

Line of regression of xx on yy

$$6x+15y=27 \quad 6x+15y=27$$

$$\Rightarrow 6x = -15y + 27 \Rightarrow 6x = -15y + 27$$

$$\Rightarrow x = -15y + 276 \Rightarrow x = -15y + 276$$

$$\Rightarrow x = -1.5y + 4.5 \Rightarrow x = -1.5y + 4.5$$

$$\therefore b_{xy} = -1.5 \therefore b_{xy} = -1.5$$

Line of regression of yy on xx

$$6x+3y=15 \quad 6x+3y=15$$

$$\Rightarrow 3y = -6x + 15 \Rightarrow 3y = -6x + 15$$

$$\Rightarrow y = -6x + 153 \Rightarrow y = -6x + 153$$

$$\Rightarrow y = -2x + 5 \Rightarrow y = -2x + 5$$

$$\therefore b_{yx} = -2 \therefore b_{yx} = -2$$

According to this assumption, $b_{xy} = -1.5$ and $b_{yx} = -2$

We know that,

$$r = \frac{b_{xy}}{b_{yx}} \therefore r = \frac{b_{xy}}{b_{yx}}$$

$$\therefore r = \frac{-1.5}{-2} \therefore r = 0.75$$

$$\therefore r = 0.75 \therefore r = 0.75$$

This assumption gives a value of $r > 1$, which is not possible. **It means our assumption is incorrect.**

Thus,

$6x+15y=27$ – Line of regression of yy on xx and $6x+3y=15$ – Line of regression of xx on yy

Line of regression of yy on xx

$$6x+15y=27 \quad 6x+15y=27$$

$$\therefore 15y = -6x + 27 \therefore 15y = -6x + 27$$

$$\therefore y = -6x + 2715 \therefore y = -6x + 2715$$

$$\therefore y = -0.4x + 1.8 \therefore y = -0.4x + 1.8$$

$$\therefore b_{yx} = -0.4 \quad \therefore b_{yx} = -0.4$$

Line of regression of xx on yy

$$6x + 3y = 15 \quad 6x + 3y = 15$$

$$\therefore 6x = -3y + 15 \quad \therefore 6x = -3y + 15$$

$$\therefore x = -3y + 156 \quad \therefore x = -3y + 156$$

$$\therefore x = -0.5y + 2.5 \quad \therefore x = -0.5y + 2.5$$

$$\therefore b_{xy} = -0.5 \quad \therefore b_{xy} = -0.5$$

Angle between the two lines of regression,

$$\theta = \tan^{-1} \left| \frac{1 - b_{xy}b_{yx}}{b_{xy} + b_{yx}} \right| \quad \theta = \tan^{-1} \left| \frac{1 - b_{xy}b_{yx}}{b_{xy} + b_{yx}} \right|$$

$$\therefore \theta = \tan^{-1} \left| \frac{1 - (-0.5)(-0.4)}{(-0.5) + (-0.4)} \right| \quad \therefore \theta = \tan^{-1} \left| \frac{1 - (-0.5)(-0.4)}{(-0.5) + (-0.4)} \right|$$

$$\therefore \theta = \tan^{-1} \left| \frac{1 - 0.2}{-0.9} \right| \quad \therefore \theta = \tan^{-1} \left| \frac{1 - 0.2}{-0.9} \right|$$

$$\therefore \theta = \tan^{-1} \left| \frac{0.8}{-0.9} \right| \quad \therefore \theta = \tan^{-1} \left| \frac{0.8}{-0.9} \right|$$

$$\therefore \theta = \tan^{-1} \left| \frac{8}{-9} \right| \quad \therefore \theta = \tan^{-1} \left| \frac{8}{-9} \right|$$

Summary

There are two lines of regression, each trying to minimise the deviations of xx and yy from their means by the method of least squares. The two lines of regression intersect at the point whose coordinates give the means of both the variables, i.e. $(\bar{x}, \bar{y})$. The formula for the angle between the two lines of regression is $\theta = \tan^{-1} \left| \frac{1 - r^2}{(1 - r^2)\sigma_x\sigma_y} \right|$. When the angle between the two lines of regression is 00 or $\pi\pi$ then there is a perfect correlation between the two variables. When the angle between the two lines of regression is $\frac{\pi}{2}$ then there is no correlation between the two variables. Thus, the angle between the two lines of regression is a connecting link between regression and correlation.

FAQs on Angle Between Two Lines of Regression

Students might be having many questions with respect to the Angle Between Two Lines of Regression. Here are a few commonly asked questions and answers.

Q.1. How do you find the angle between two regression lines?

Ans: You can find the angle between the two lines of regression by using the formula

$$\theta = \tan^{-1} \left| \frac{1 - r^2}{(1 - r^2)\sigma_x\sigma_y} \right| \quad \theta = \tan^{-1} \left| \frac{1 - r^2}{(1 - r^2)\sigma_x\sigma_y} \right|$$

Here,

$\sigma_x = \sigma_x$ = Standard deviation of the values of xx from $\bar{x}$

$\sigma_y = \sigma_y$ = Standard deviation of the values of yy from $\bar{y}$

$r = r$ = Correlation coefficient

Q.2. What are the two lines of regression?

Ans: The two lines of regressions are a line of regression of xx on yy and a line of regression of yy on xx .

- Line of regression of xx on yy – It establishes a relationship between xx and yy . It is used to find the unknown value of the dependent variable xx for a known value of the independent variable yy .
- Line of regression of yy on xx – It establishes a relationship between yy and xx . It is used to find the unknown value of the dependent variable yy for a known value of the independent variable xx .

Q.3. Under what condition will the angle between two regression lines become zero?

Ans: The angle between the two regression lines becomes zero when the two variables are in perfect correlation, either positive or negative.

Q.4. When the correlation coefficient increases from 00 to 11, how does the angle between the regression lines diminish?

Ans: When the correlation coefficient increases from 00 to 11, the angle between the regression lines diminishes from $\pi/2$ to 00.

Q.5. Why there are two regression lines? Write the properties of regression lines.

Ans: There are two lines of regression each trying to minimise the squares of deviations of xx and yy from their means by the method of least squares.

Line of regression of xx on yy – It minimises the deviations of squares of values of xx from its mean $\bar{x}$ parallel to the XX -axis.

Line of regression of yy on xx – It minimises the deviations of squares of values of yy from its mean $\bar{y}$ parallel to the YY -axis.

Linear regression attempts to model the relationship between two variables by fitting a linear equation to observed data. One variable is considered to be an explanatory variable, and the other is considered to be a dependent variable. For example, a modeler might want to relate the weights of individuals to their heights using a linear regression model.

Before attempting to fit a linear model to observed data, a modeler should first determine whether or not there is a relationship between the variables of interest. This does not necessarily imply that one variable *causes* the other (for example, higher SAT scores do not *cause* higher college grades), but that there is some significant association between the two variables. A scatterplot can be a helpful tool in determining the strength of the relationship between two variables. If there appears to be no association between the proposed explanatory and dependent variables (i.e., the scatterplot does not indicate any increasing or decreasing trends), then fitting a linear regression model to the data probably will not provide a useful model. A valuable numerical measure of association between two variables is the correlation coefficient, which is a value between -1 and 1 indicating the strength of the association of the observed data for the two variables.

A linear regression line has an equation of the form $Y = a + bX$, where X is the explanatory variable and Y is the dependent variable. The slope of the line is b , and a is the intercept (the value of y when $x = 0$).

Multiple Regression

Simple linear regression is a technique for predicting the value of a dependent variable, based on the value of a single independent variable. Sometimes, you only need one relevant independent variable to make an accurate prediction.

More often, however, the prediction is better when you use two or more independent variables. Multiple regression is a technique for predicting the value of a dependent variable, based on the values of two or more independent variables.

The Regression Equation

This is a tutorial about *linear* regression, so our focus is on *linear* relationships between variables. The regression equation that expresses the linear relationships between a single dependent variable and one or more independent variables is:

$$\hat{y} = b_0 + b_1x_1 + b_2x_2 + \dots + b_{k-1}x_{k-1} + b_kx_k$$

In this equation, $\hat{y}$ is the *predicted* value of the dependent variable. Values of the k independent variables are denoted by $x_1, x_2, x_3, \dots, x_k$.

And finally, we have the b 's - $b_0, b_1, b_2, \dots, b_k$. The b 's are constants, called regression coefficients. Values are assigned to the b 's based on the principle of least squares.

What is the Principle of Least Squares?

In multiple regression, the deviation of the actual value for a dependent variable from its predicted value is called the residual. The residual (e) for a single observation i is:

$$e_i = y_i - \hat{y}_i = y_i - (b_0 + b_1x_{1i} + b_2x_{2i} + \dots + b_kx_{ki})$$

Assume that the set of data consists of n observations. The principle of least squares requires that the sum of squared residuals for all n observations be minimized. That is, we want the following value to be as small as possible:

$$\sum [y_i - (b_0 + b_1x_{1i} + b_2x_{2i} + \dots + b_kx_{ki})]^2$$

Regression analysis requires that the values of $b_0, b_1, \dots, b_k$ be defined to minimize the sum of the squared residuals. When we assign values to regression coefficients in this way, we are following the principle of least squares.

Normal Equations for Simple Regression

Finding the right values for regression coefficients (i.e., values that satisfy a least squares criterion) involves solving a set of linear equations. These equations can be derived using calculus, and they are called normal equations.

To illustrate the use of normal equations, let's look at simple linear regression - regression with one dependent variable (y) and one independent variable (x). With simple linear regression, the regression equation is:

$$\hat{y} = b_0 + b_1x$$

The normal equations for simple linear regression are:

$$\sum y_i = nb_0 + b_1(\sum x_i)$$

$$\sum x_i y_i = b_0(\sum x_i) + b_1(\sum x_i^2)$$

Here, we have two equations with two unknowns. The unknowns are the regression coefficients b_0 and b_1 . Using ordinary algebra, we can solve for b_0 and b_1 . The result is:

$$b_1 = \frac{\sum [(x_i - \bar{x})(y_i - \bar{y})]}{\sum [(x_i - \bar{x})^2]}$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

where $\bar{x}$ is the mean x score, and $\bar{y}$ is the mean y score. Note that these are the same equations that we presented in a previous lesson, when we introduced the topic of simple linear regression.

The use of normal equations to assign values to regression coefficients becomes more complicated when there are two or more independent variables. We'll tackle that challenge in the next lesson.

Regression is used in statistical modeling and it basically tells us the relationship between variables and their movement in the future. Apart from statistical methods like standard deviation, regression, correlation. The regression analysis is the most widely and commonly accepted measure to measure the variance in the industry. These relationships are seldom exact because there is variation caused by many variables, not just the variables being studied. The method is widely used in the industry for predictive modeling and forecasting measures.

Regression tells us the relationship of the independent variable on the dependent variable and to explore the forms of these relationships.

The formula for Regression Analysis –

$$Y = a + bX + \epsilon$$

- Y = Stands for the dependent variable
- X = Stands for an independent variable
- a = Stands for the intercept
- b = Stands for the slope
- ϵ = Stands for the error term

The formula for intercept “a” and the slope “b” can be calculated as per below.

$$a = \frac{(\sum y)(\sum x^2) - (\sum x)(\sum xy)}{n(\sum x^2) - (\sum x)^2}$$

$$b = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$$

Regression analysis is one of the most powerful multivariate statistical technique as the user can interpret parameters the slope and the intercept of the functions that link with two or more variables in a given set of data.

There are two types of regression multilinear regression and simple linear regression. The simple linear regression is explained and is the same as $Y = a + bX_1 + cX_2 + dX_3 + \epsilon$

Where,

- Y – Dependent variable
- X_1, X_2, X_3 – Independent (explanatory) variables
- a – Intercept

- b, c, d – Slopes
- ϵ – Residual (error)

Example #1

Consider the following two variables x and y , you are required to do the calculation of the regression.

	A	B	C
1	X	Y	
2	34.86	43.04	
3	42.58	51.88	
4	71.73	88.55	
5	110.77	130.69	
6	259.95	314.17	
7			

Solution:

Using the above formula, we can do the calculation of [linear regression in excel](#) as follows.

	A	B	C	D	E
1	X	Y	XY	X^2	Y^2
2	34.86	43.04	1500.3744	1215.2196	1852.4416
3	42.58	51.88	2209.0504	1813.0564	2691.5344
4	71.73	88.55	6351.6915	5145.1929	7841.1025
5	110.77	130.69	14476.5313	12269.9929	17079.8761
6	259.95	314.17	81668.4915	67574.0025	98702.7889
7	$(\sum x) = 519.89$	$(\sum y) = 628.33$	$(\sum XY) = 106206.14$	$(\sum X^2) = 88017.46$	$(\sum Y^2) = 128167.74$
8					

We have all the values in the above table with $n = 5$.

Now, first, calculate the intercept and slope for the regression.

Calculation of Intercept is as follows,

C9		f_x	$=((B7*D7)-(A7*C7))/(5*D7-A7^2)$		
	A	B	C	D	
1	X	Y	XY	X^2	
2	34.86	43.04	1500.3744	1215.2196	1852.4416
3	42.58	51.88	2209.0504	1813.0564	2691.5344
4	71.73	88.55	6351.6915	5145.1929	7841.1025
5	110.77	130.69	14476.5313	12269.9929	17079.8761
6	259.95	314.17	81668.4915	67574.0025	98702.7889
7	519.89	628.33	106206.14	88017.46	128167.74
8					
9	a (intercept)		0.52		
10					

$$a = \frac{(\sum y)(\sum x^2) - (\sum x)(\sum xy)}{n(\sum x^2) - (\sum x)^2}$$

$$a = (628.33 * 88,017.46) - (519.89 * 106,206.14) / 5 * 88,017.46 - (519.89)^2$$

$$a = 0.52$$

Calculation of Slope is as follows,

C9		f_x	$=((5*C7)-(A7*B7))/(5*D7-A7^2)$		
	A	B	C	D	
1	X	Y	XY	X^2	
2	34.86	43.04	1500.3744	1215.2196	1852.4416
3	42.58	51.88	2209.0504	1813.0564	2691.5344
4	71.73	88.55	6351.6915	5145.1929	7841.1025
5	110.77	130.69	14476.5313	12269.9929	17079.8761
6	259.95	314.17	81668.4915	67574.0025	98702.7889
7	519.89	628.33	106206.14	88017.46	128167.74
8					
9	b (slope)		1.20		
10					

$$b = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$$

$$b = (5 * 106,206.14) - (519.89 * 628.33) / (5 * 88,017.46) - (519.89)^2$$

$$b = 1.20$$

Let's now input the values in the regression formula to get regression.

Hence the regression line $Y = 0.52 + 1.20 * X$

References:

Advance Engineering Mathematics, Chandrika Prasad, Standard Publishers and Distributors